

Connections

➡ Building on

- Acidity and basicity **ch8**
- Carbonyl chemistry **ch10 & ch11**
- Stereochemistry **ch14**
- Conformational analysis **ch16**
- Enolate chemistry and synthesis **ch25 & ch26**
- Sulfur chemistry **ch27**
- Heterocycles **ch29 & ch30**
- Asymmetric synthesis **ch41**

Arriving at

- Nucleic acids store information for the synthesis of proteins
- Modified nucleosides can be used as antiviral drugs
- Proteins catalyse reactions and provide structure
- Other amino acid derivatives act as methylating and reducing agents
- Sugars store energy, enable recognition, and protect sensitive functional groups
- How to make and manipulate sugars and their derivatives
- Lipids form the basis of membrane structures
- The main sorts of natural products are alkaloids, polyketides, terpenes, and steroids
- Alkaloids are amines made from amino acids
- Fatty acids are built up from acetyl CoA and malonyl CoA subunits

➡ Looking forward to

- Three more comprehensive web chapters:
 - The chemistry of life
 - Mechanisms in biological chemistry
 - Natural products
- Organic chemistry today **ch43**

Primary metabolism

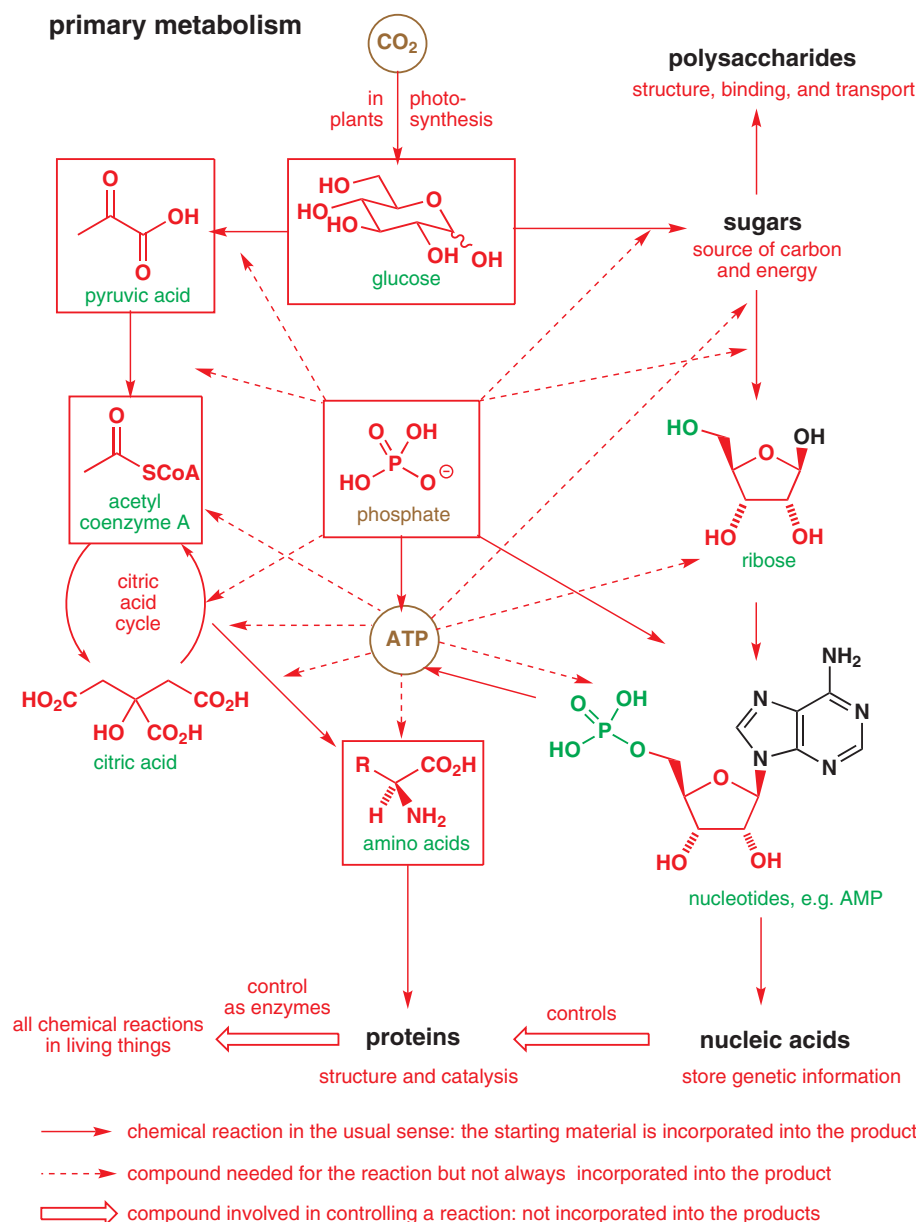
Secondary metabolism is, by contrast, chemistry less fundamental to the workings of life and restricted to smaller groups of organisms. Later in this chapter you will meet alkaloids produced by some plants and terpenes produced by others. Humans produce neither of these, but we do make steroids, as do other animals (and a few plants). All of these molecules are the products of secondary metabolism.

Life runs on chemistry, and the chemical side of biology is fascinating for that reason alone. It is humbling to realize that the same molecules are present in all living things, from the simplest single-cell creatures to ourselves. Nucleic acids contain the genetic information of every organism, and they control the synthesis of proteins. Proteins are partly structural—as in connective tissue—and partly functional—as in enzymes, the catalysts for biological reactions. Sugars and lipids used to be thought of as the poor relations of the other two, storing energy and building membranes, but it is now clear that they also have a vital part to play in recognition and transport.

The chemistry common to all living things is known as *primary metabolism* and the chart overleaf shows the molecules of primary metabolism and the connections between them, and needs some explanation. It shows a simplified relationship between the key structures (emphasized in large black type). It shows their origins—from CO₂ in the first instance—and picks out some important intermediates. Glucose, pyruvic acid, citric acid, acetyl coenzyme A (acetyl CoA), and ribose are players on the centre stage of metabolism and are built into many

Online support. The icon  in the margin indicates that accompanying interactive resources are provided online to help your understanding: just type www.chemtube3d.com/clayden/123 into your browser, replacing **123** with the number of the page where you see the icon. For pages linking to more than one resource, type **123-1**, **123-2** etc. (replacing **123** with the page number) for access to successive links.

important biological molecules. Use this chart to keep track of the relationships between the molecules of metabolism as you develop a more detailed understanding of them. We start with nucleic acids.

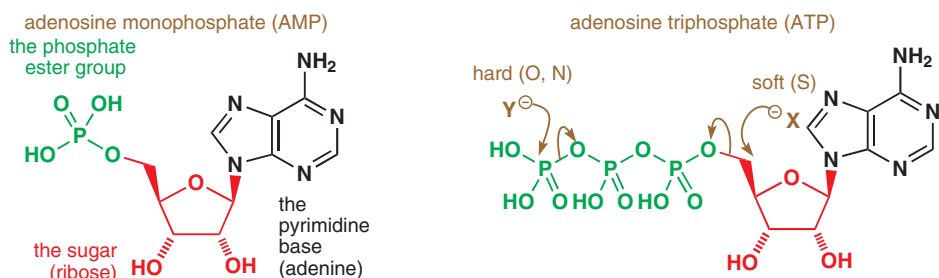


Life begins with nucleic acids

Nucleic acids store genetic information. They are polymers whose building blocks (monomers) are the nucleotides, themselves made of three parts—a heterocyclic base, a sugar, and a phosphate ester. In the example below, adenine is the base (shown in black), adenosine is the nucleoside (base and sugar), and the nucleotide is the whole molecule (base + sugar + phosphate). This nucleotide is called AMP—adenosine monophosphate. Phosphates are key compounds in nature because they form useful stable linkages between molecules and can also be built up into reactive molecules by simply multiplying the number of phosphate residues. The most important of these nucleotides is also one of the most important molecules in nature—adenosine triphosphate or ATP.

Nucleosides and nucleotides

A nucleoside differs from a nucleotide in lacking the phosphate—a nucleoside is just a base and a sugar.



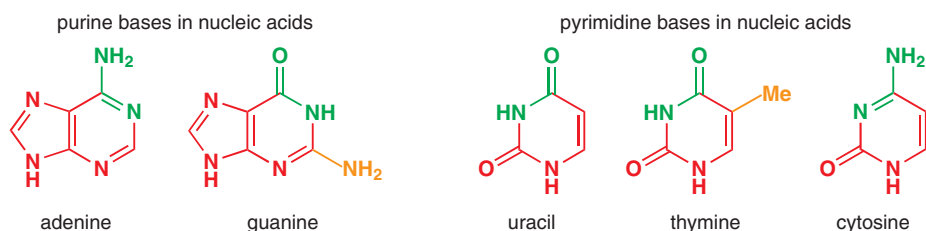
ATP is a highly reactive molecule because phosphates are stable anions and good leaving groups. It can be attacked by hard nucleophiles at a phosphate group (usually the end one) or by soft nucleophiles at the CH₂ group on the sugar. When a new reaction is initiated in nature, very often the first step is a reaction with ATP to make the compound more reactive. This is rather like our use of TsCl to make alcohols more reactive or converting acids to acid chlorides to make them more reactive.

There are five heterocyclic bases in DNA and RNA

Nucleic acids are made up of a selection of five bases, two sugars, and the phosphate group. The bases are monocyclic pyrimidines or bicyclic purines and are all aromatic.

- There are only two purine bases found in nucleic acids: adenine (A), which we have already met, and guanine (G).
- The three pyrimidine bases are simpler: uracil (U), thymine (T), and cytosine (C). Cytosine is found in DNA and RNA, uracil in RNA only, and thymine in DNA only.

The coloured parts of the molecules below emphasize the characteristic features of the bases.



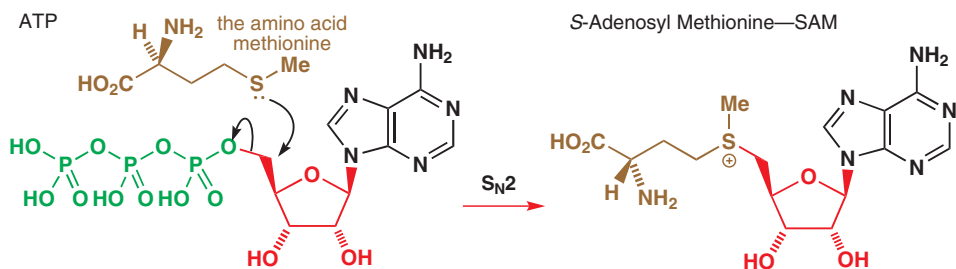
■ You met pyrimidines in Chapter 29 and learned how to make them in Chapter 30, but the purine ring system may be new to you. Make sure you can find the six (or ten) electrons making these compounds aromatic. You may need to draw delocalized structures, especially for U, T, and G.

The stimulants in tea and coffee are methylated purines

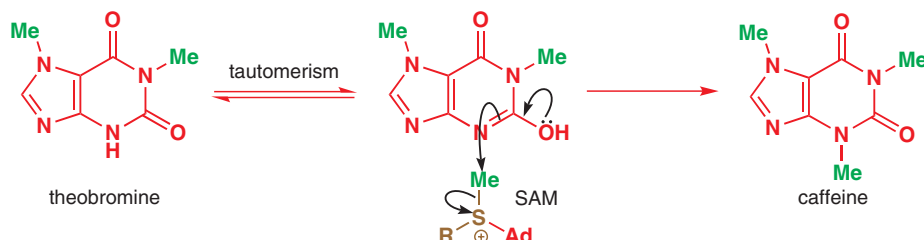
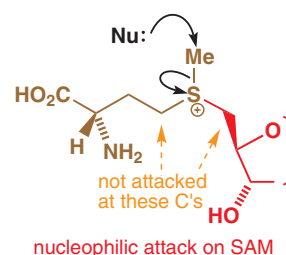
An important stimulant for many is a fully methylated purine present in tea and coffee—caffeine. Caffeine is a crystalline substance easily extracted from coffee or tea with organic solvents. It is extracted industrially with supercritical CO₂ (or, if you prefer, 'nature's effervescence') to make decaffeinated tea and coffee.



If we, as chemists, were to add those methyl groups we would choose to use a reagent such as methyl iodide, but nature uses a much more complicated molecule. There is a great deal of methylating going on in living things—and the methyl groups are usually added by (S)-adenosyl methionine (or SAM), formed by reaction of methionine with ATP. This is a good reaction because sulfur is a good soft nucleophile, triphosphate is a good leaving group, and substitution at primary carbon is easy.



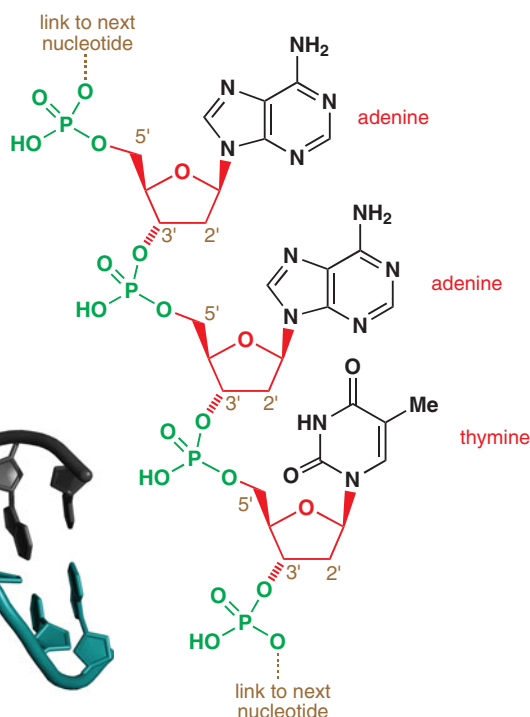
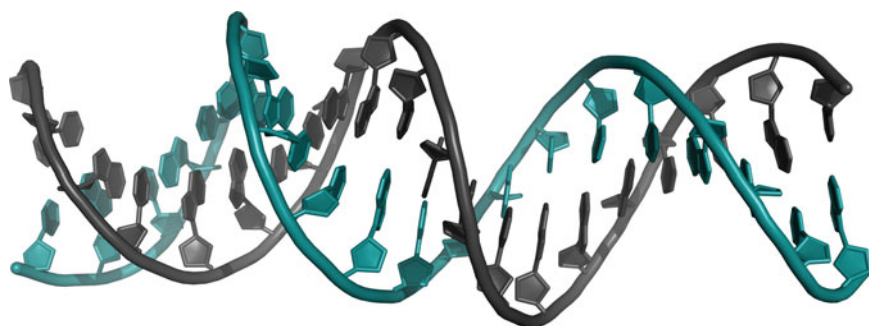
SAM is a sulfonium salt and could be attacked by nucleophiles at three different carbon atoms. Two are primary centres—good for S_N2 reactions—but the third is the methyl group, which is even better. Many nucleophiles attack SAM in this way. In the coffee plant, theobromine (a purine also found in cocoa) is converted into caffeine with a molecule of SAM. The methylation occurs on nitrogen partly because this preserves both the aromatic ring and the amide functionality and also because the enzyme involved brings the two molecules together in the right orientation for *N*-methylation.



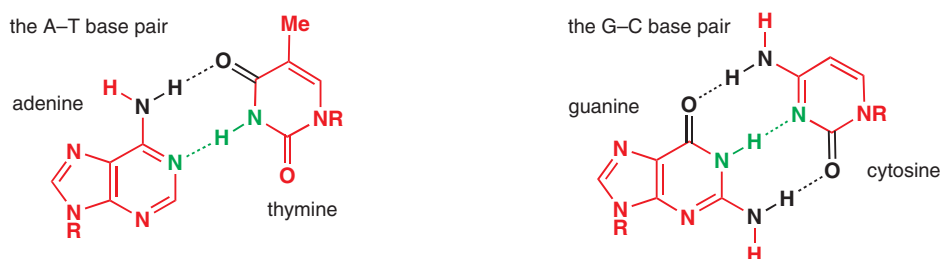
At this point we should just point out something that it's easy to forget: there is *only one chemistry*. There is no magic in biological chemistry, and nature uses the same chemical principles as we do in the chemical laboratory. All the mechanisms that you have studied so far will help you to draw mechanisms for biological reactions and most reactions that you have met have their counterparts in nature. The difference is that nature is very, very good at chemistry, and we humans are only just learning. We still do much more sophisticated reactions *inside* our bodies without thinking about them than we can do *outside* our bodies with all the most powerful ideas available to us in the 21st century.

Nucleic acids exist in a double helix

One of the most important discoveries of modern science was the elucidation of the structures of DNA and RNA as the famous double helix by Watson and Crick in 1953. They realized that the basic structure of base–sugar–phosphate was ideal for a three-dimensional coil. The structure of a small part of DNA is shown on the right. Notice that the 2' (pronounced 'two prime') position on the ribose ring is vacant. There is no hydroxyl group there: that is why it is called *deoxyribonucleic acid*. The nucleotides link the two remaining OH groups on the ribose ring and these are called the 3'- and 5'-positions. This piece of DNA has three nucleotides (adenine, adenine, and thymine) and so would be called –AAT– for short.



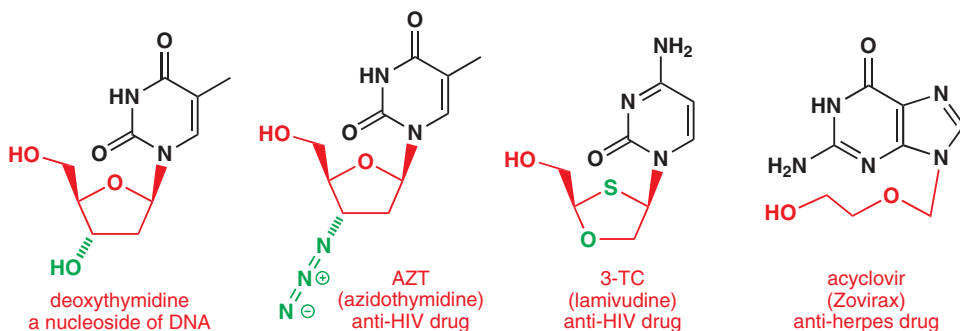
Each polymeric strand of DNA coils up into a helix and is bonded to another strand by hydrogen bonds between the bases. Each base pairs up specifically with another base—adenine with thymine (A–T) and guanine with cytosine (G–C)—like this.



There is quite a lot to notice about these structures. Each purine (A or G) is bonded specifically to one pyrimidine (T or C) by two or by three hydrogen bonds. The hydrogen bonds are of two kinds: one links an amine to a carbonyl group (black in the diagram) and one links an amine to an imine (green in the diagram). A purine has to pair with a pyrimidine because only the combination of larger purine and smaller pyrimidine bridges the gap between the nucleic acid coils. Look back at the green and orange parts of the structures on p. 1136 and you will see that only one hydrogen bond pairing pattern can work. In this way, each nucleotide reliably recognizes another and reliably pairs with its partner. The short strand of DNA above (–AAT–) would pair reliably with –TTA–.

HIV and AIDS are treated with modified nucleosides

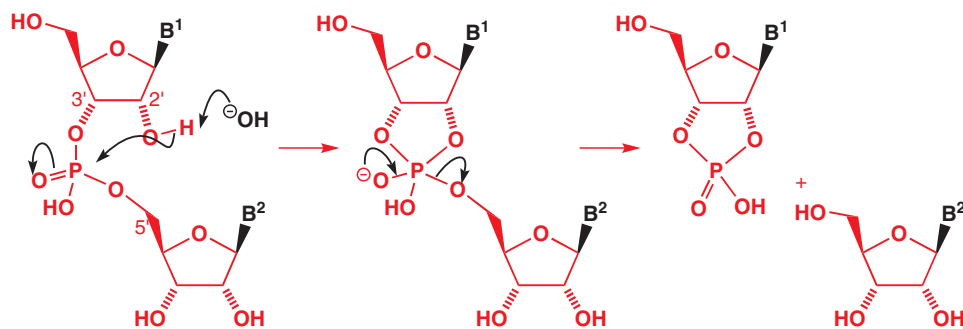
Modified nucleosides are among the best antiviral compounds. The anti-HIV drug AZT (zidovudine) is a slightly modified DNA nucleoside (3'-azidothymidine). It has an azide at C3' instead of the hydroxyl group in the natural nucleoside. A more radically modified nucleoside 3-TC (lamivudine) is active against AZT-resistant viruses. This drug is based on cytosine with the sugar replaced by a different heterocycle, although it is recognizably similar, especially in the stereochemistry. Acyclovir (Zovirax), the cold sore (herpes) treatment, is a modified guanosine in which only a ghost of the sugar remains. There is no ring at all and no stereochemistry.



Cyclic nucleosides and stereochemistry

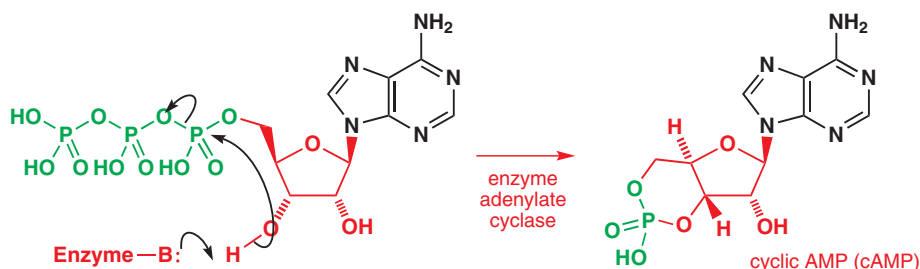
DNA is more stable than RNA because its sugars lack the 2' hydroxyl groups. In ribonucleic acids, the fact that the 2'- and 3'-OH groups are on the same side of the ring makes alkaline hydrolysis exceptionally rapid by intramolecular nucleophilic catalysis.

■ The substituents B¹ and B² represent any purine or pyrimidine base.



The base removes a proton from the 2'-OH group, which cyclizes on to the phosphate link—possible only if the ring fusion is *cis*. The next reaction involves breakdown of the pentacoordinate phosphorus intermediate to give a cyclic phosphate. One nucleoside is released by this reaction and the second follows when the cyclic phosphate is itself cleaved by base.

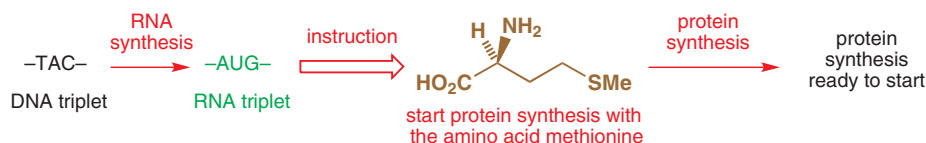
Another cyclic phosphate that can be formed from a nucleotide is important as a biological messenger that helps to control such processes as blood clotting and acid secretion in the stomach. It is cyclic AMP (cAMP), formed enzymatically from ATP by nucleophilic displacement of pyrophosphate by the 3'-OH group.



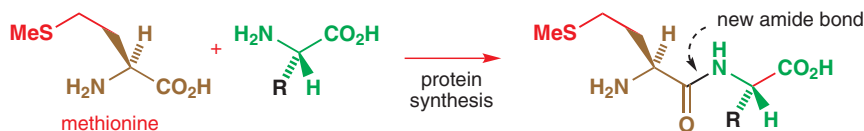
■ Note that cAMP has a *trans* 6,5-fused ring junction.

Proteins are made of amino acids

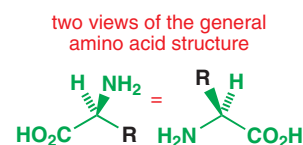
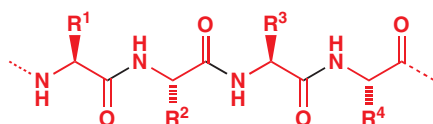
DNA encodes the information needed to make proteins in the form of triplets of bases (codons), for example thymine–adenine–cytosine (TAC) in the diagram below. As RNA is synthesized from DNA, these are turned into complementary codons (in the example below, AUG) by pairing up the bases as shown on p. 1138. This RNA forms the instructions for protein synthesis by the ribosome—perhaps the most elaborate molecular structure in the known universe. Each codon of the RNA chain tells the ribosome to add a specific amino acid to the growing protein. For example, the codon AUG indicates methionine, which we met as a component of SAM. Methionine is a typical amino acid of the kind present in proteins, but is also the starter unit of all proteins.



The next codon of RNA directs the ribosome to add the next amino acid, linked to the previous one in the chain by an amide bond. Amino acids used to make proteins have the same basic structure and stereochemistry, shown in the margin, and differ only in the group R.



The process continues as more amino acids are added in turn to the right-hand end of the growing molecule. A section of the final protein might look like the structure below. The skeleton of the protein zig-zags up and down in the usual way; the amide bonds (shown in black) are rigid because of the amide conjugation and are held in the shape shown.



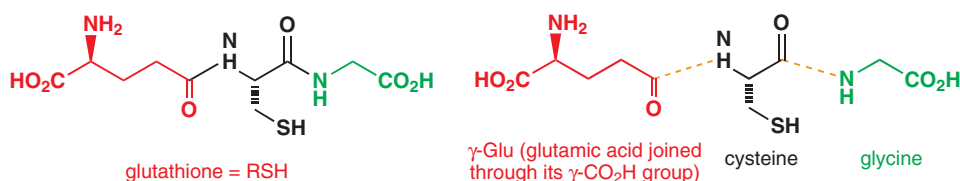
➡ There is a list of the naturally occurring amino acids in Chapter 23 (p. 554), where we discussed the *laboratory* synthesis of peptides.

■ Much of the function of enzymes and other proteins derives from their detailed folded conformation, discussion of which is beyond the scope of this book.

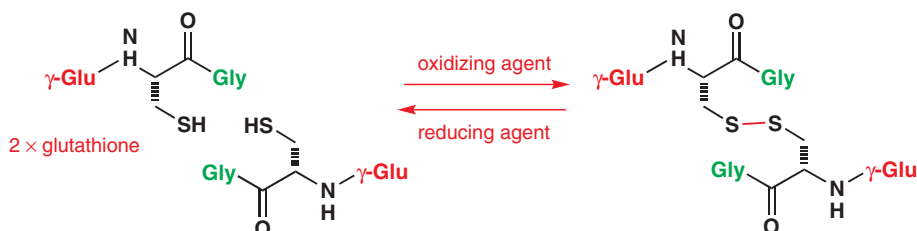
Amino acids combine to form peptides and proteins

In nature, the amino acids are combined to give proteins with hundreds or even thousands of amino acids in each one. Small assemblies of amino acids are known as **peptides** and the amide bond that links them is called a **peptide bond**.

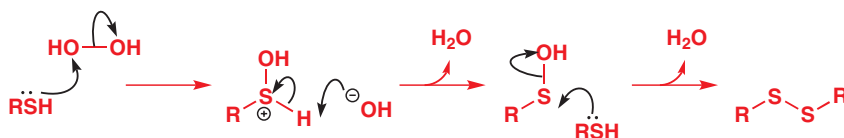
An important tripeptide is **glutathione**, present in the tissues of both animals and plants. Glutathione is the ‘universal thiol’ that removes dangerous oxidizing agents by allowing itself to be oxidized to a disulfide. Glutathione is, however, not quite a typical tripeptide. The left-hand amino acid is normal glutamic acid but it is joined to the next amino acid through its γ -CO₂H group instead of the more normal α -CO₂H group. The middle amino acid is the vital one for the function—cysteine with a free SH group. The C-terminal acid is glycine.



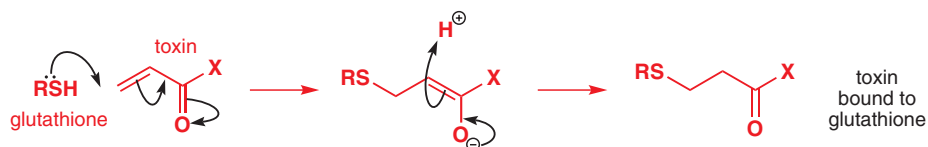
Thiols are easily oxidized to disulfides and glutathione sacrifices itself if it meets an oxidizing agent. The oxidized form of glutathione can later be converted back to the thiol by reduction with NADH, which you will meet later in this chapter.



With a stray oxidizing agent such as a peroxide, say H₂O₂, the mechanism below shows how this can be reduced to water as glutathione (represented as RSH) is oxidized to a disulfide.

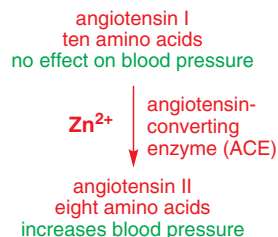


Glutathione also detoxifies some of the compounds we described earlier in this book as dangerous carcinogens such as Michael acceptors and 2,4-dinitrohalobenzenes. The thiol acts as a nucleophile, inactivating the electrophiles. Covalently bound to glutathione they are harmless and can be excreted. More glutathione will be synthesized from glutamic acid, cysteine, and glycine to replace that which is lost.



➡ See, for example, the discussion in Chapter 22, p. 516.

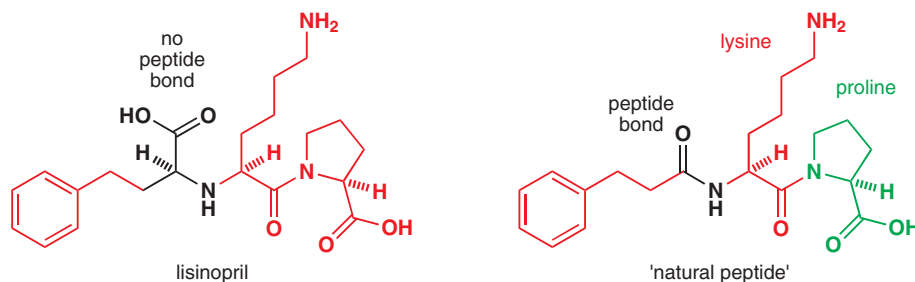
➡ You saw some examples in Chapter 23.



Some short peptides, of around ten amino acids, are hormones. Angiotensin II, for example, is a peptide that causes blood pressure to rise—a very necessary thing in some situations but too much and too often leads to heart attacks and strokes.

Angiotensin-converting enzyme (ACE) is the zinc-dependent enzyme that cleaves two amino acids from the end of angiotensin I to give angiotensin II, and ACE inhibitors are used as treatment for high blood pressure because they inhibit this enzyme. Lisinopril is an example: it is a dipeptide mimic, having two natural amino acids and something else. The ‘something else’ is the left-hand part of the molecule, linked to the dipeptide (Lys–Pro) through an amine and not

by an amide bond. This stops enzymes from hydrolysing the molecule. Lisinopril binds to ACE because it is *like* a natural dipeptide but it inhibits it because it is *not* a natural dipeptide. Many people are alive today because of this simple deception practised on an enzyme.



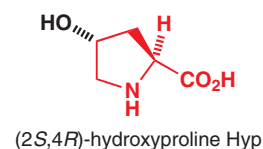
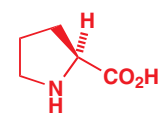
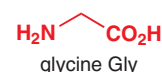
Structural proteins must be tough and flexible

In contrast with the functional enzymes, proteins such as collagen are purely structural. Collagen is the tough protein of tendons and connective tissue, and is present in skin, bone, and teeth. It contains large amounts of glycine (every third amino acid is glycine), proline, and hydroxyproline (again about a third of the amino acids are either Pro or Hyp).

Hydroxyproline is a specialized amino acid that appears almost nowhere else and, along with proline, it establishes a very strong triply coiled structure for collagen. The glycine is necessary as there is no room in the triple coil for any larger amino acid. Functionalized amino acids are rare in collagen.

Hydroxyproline and scurvy

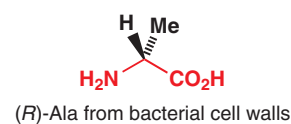
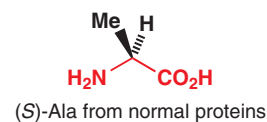
Hydroxyproline is a very unusual amino acid. It is not incorporated into the growing protein chain when collagen is synthesized—instead the collagen molecule is assembled with Pro where Hyp is needed. Once the protein is complete, some of the proline residues are oxidized to hydroxyproline. This oxidation requires vitamin C, and without it collagen cannot be formed. This is why vitamin C deficiency causes scurvy—the symptoms of scurvy suffered by 18th-century sailors (loose teeth, sores, and blisters) were caused by the inability to make collagen.



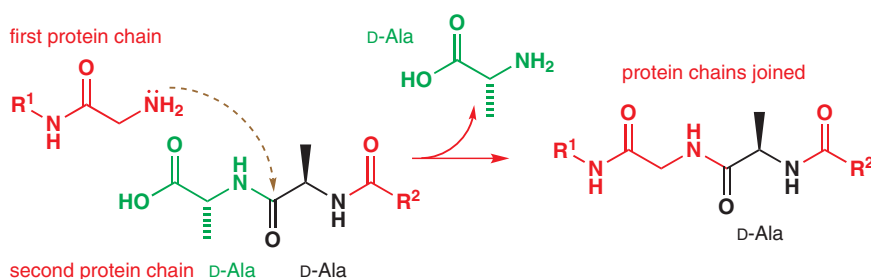
Antibiotics exploit the special chemistry of bacteria

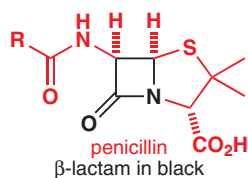
We have repeatedly emphasized that all life has very similar chemistry. From the biochemical point of view the most important division is that separating *prokaryotes* from *eukaryotes*. Prokaryotes, which include bacteria, evolved first and have simple cells with no nucleus. Eukaryotes, which include plants, mammals, and all other multicellular creatures, evolved later and have more complex cells, including nuclei. Even then, much of the biochemistry on both sides of the divide is the same.

When medicinal chemists are looking for ways to attack bacteria, one approach is to interfere with chemistry carried out by prokaryotes but not by us. The most famous of these attacks is aimed at the construction of the cell walls of some bacteria that contain 'unnatural' (*R*)- (or *D*-) amino acids. Bacterial cell walls are made from glycopeptides of an unusual kind. Polysaccharide chains are cross-linked with short peptides containing (*R*)-alanine (*D*-Ala). Before they are linked up, one chain ends with a glycine molecule and the other with *D*-Ala–*D*-Ala. In the final step in the cell wall synthesis, the glycine attacks the *D*-Ala–*D*-Ala sequence to form a new peptide bond by displacing one *D*-Ala residue.

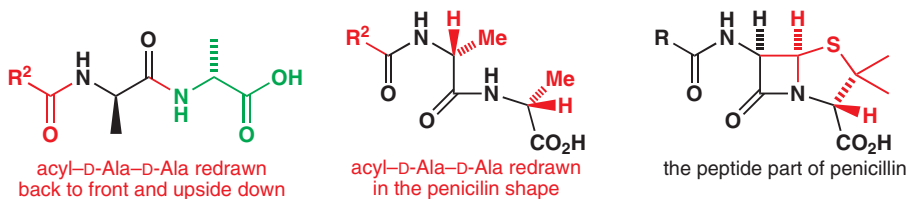


■ The reason bacteria have evolved to use these 'unnatural' *D*-amino acids in their cell walls is to protect them against the enzymes in animals and plants, which cannot digest proteins containing *D*-amino acids.



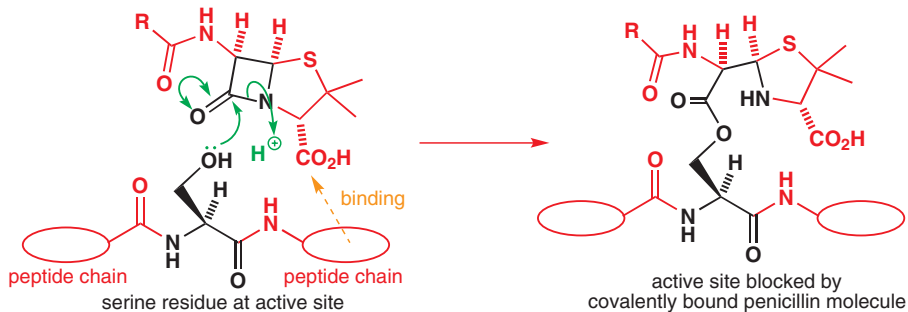


The antibiotic penicillin works by interfering with this step—although this was not even suspected when penicillin was discovered. Penicillin inhibits the enzyme that catalyses the D-Ala transfer in a very specific way. It first binds specifically to the enzyme (so it must be a mimic of the natural substrate) and it then reacts with the enzyme and inactivates it by blocking a vital OH group at the active site. If we emphasize the peptide nature of penicillin and compare it with D-Ala–D-Ala, the mimicry may become clearer.



Penicillin imitates D-Ala and binds to the active site of the enzyme, encouraging the OH group of a serine residue to attack the reactive strained β-lactam. This same OH group of the same serine residue would normally be the catalyst for the D-Ala–D-Ala cleavage used in the building of the bacterial cell wall. The reaction with penicillin ‘protects’ the serine and irreversibly inhibits the enzyme. The bacterial cell walls cannot be completed, and the bacterial cells literally burst under the pressure of their contents. Penicillin does not kill bacteria whose cell walls are already complete but it does prevent new bacteria being formed.

Our current last line of defence against bacteria resistant to penicillin, and other antibiotics, is vancomycin. Vancomycin works by binding to the D-Ala–D-Ala sequences of the bacterial cell wall.



Sugars—just energy sources?

Sugars are the building blocks of carbohydrates. They used to be thought of as essential but rather dull molecules whose function was principally the (admittedly useful) storage of energy. In fact they have much more interesting and varied roles than that. We have already noted that ribose plays an intimate role in DNA and RNA structure and function. Sugars are also often found in intimate association with proteins and are involved in recognition and adhesion processes.

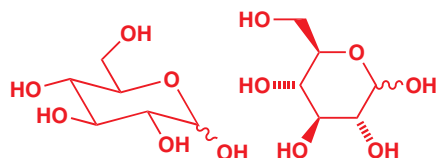
Here are two examples. How does a sperm recognize the egg and penetrate its wall? Recognition of a carbohydrate attached to the membrane of the egg was the first event in all of our lives. And how does a virus get inside a cell? Here again, the recognition process involves specific carbohydrates. One of the ways in which AIDS is being tackled with some success is by a combination of the antiviral drugs we met earlier in this chapter with HIV protease inhibitor drugs, which aim to prevent recognition and penetration of cells by HIV.

Sugars normally exist in cyclic forms with much stereochemistry

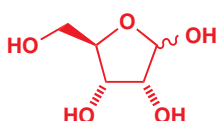
The most important sugar is glucose. It has a saturated six-membered ring containing oxygen and it is best drawn in a chair conformation with nearly all the substituents equatorial. It can also be drawn as a flat configurational diagram. We have already met one sugar in this chapter, ribose, because it was part of the structure of nucleic acids. This sugar is a five-membered saturated oxygen heterocycle with many OH groups. Indeed, you can define a sugar as an oxygen heterocycle with every carbon atom bearing an oxygen-based functional group—usually OH, but alternatively C=O.

➡ There is more about the development of drugs for treating HIV in Chapter 43.

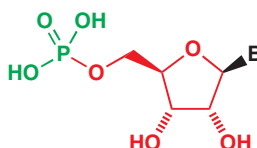
two representations of glucose



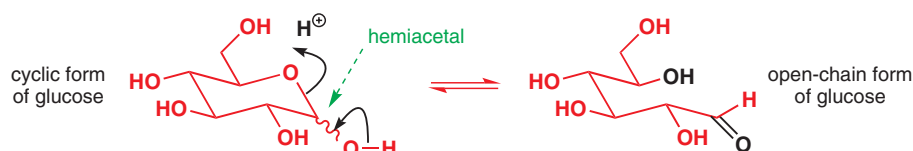
ribose



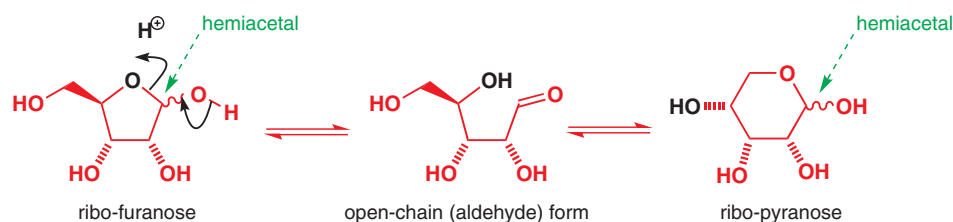
a ribonucleotide



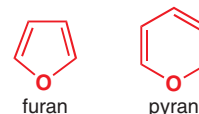
The drawings of glucose and ribose show a number of stereogenic centres, with one centre undefined—an OH group shown with a wavy bond. This is because one centre in both sugars is a hemiacetal and therefore the molecule is in equilibrium with an open-chain hydroxy-aldehyde. For glucose, the open-chain form is this.



When the ring closes again, any of the OH groups could cyclize on to the aldehyde but there is no real competition—the six-membered ring is more stable than any of the alternatives (which could have three-, four-, five-, or seven-membered rings—check for yourself). However, with ribose there is a reasonable alternative.

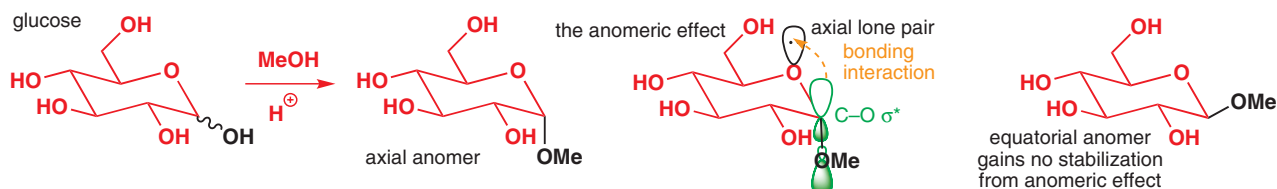


The most important sugars may exist in an open-chain form, as a five-membered oxygen heterocycle (called a *furanose*, after the five-membered aromatic compound furan) or a six-membered oxygen heterocycle (called a *pyranose*, after the six-membered pyran). Glucose prefers the pyranose structure; ribose prefers the furanose structure.



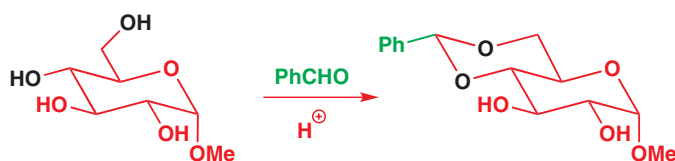
Sugars can be fixed in one shape by acetal formation

The simplest way to fix glucose in the pyranose form is to trap it as an acetal. Acid-catalysed condensation with an alcohol, methanol, for example, gives an acetal and, remarkably, the acetal has an *axial* OR group. Acetal formation is under thermodynamic control (Chapter 11) so the axial compound must be the more stable. This is because of the anomeric effect—so-called because this C atom is called the anomeric position and the acetal diastereoisomers are called anomers. The effect is a bonding interaction between the axial lone pair on the oxygen atom in the ring and the σ^* orbital of the OMe group.



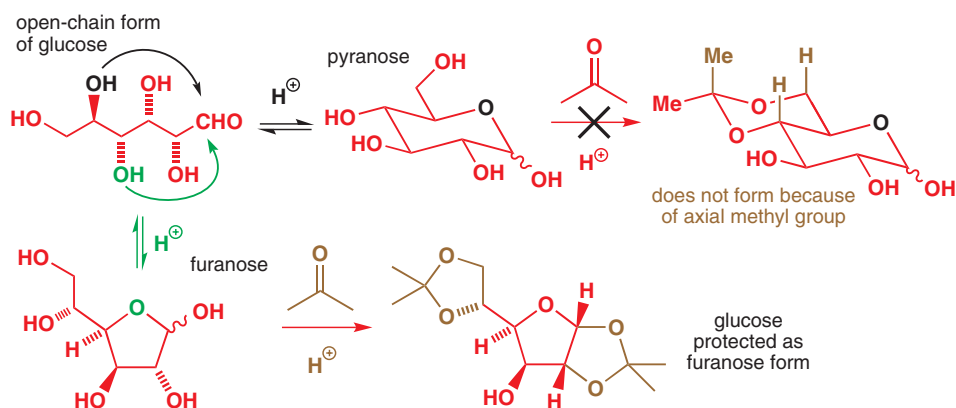
➡ The anomeric effect was discussed in Chapter 31, and you should check that you can still write down the mechanism of acetal formation you learned in Chapter 11.

The formation of acetals allows a remarkable degree of control over the chemistry of sugars. Apart from the simple glucoside acetal we have just seen, there are three important acetals worth understanding because of the way in which they illustrate stereoelectronic effects—the interplay of stereochemistry and mechanism. If we make an acetal from methyl glucoside and benzaldehyde, we get a single compound as a single stereoisomer.



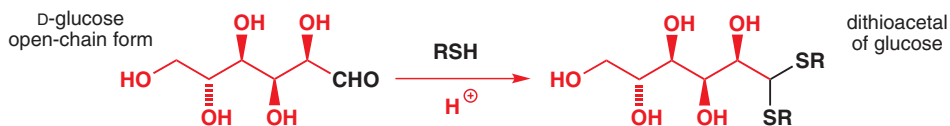
The new acetal could have been formed between any of the adjacent OH groups in the starting material but it chose the only pair (the black OH groups) which give a six-membered ring. The stereochemistry of glucose is such that the new six-membered ring is *trans*-fused to the old so that a beautifully stable all-chair bicyclic structure results, with the phenyl group in an equatorial position in the new chair acetal ring. Acetal formation is under thermodynamic control and this product is the most stable possible acetal.

Acetal formation from sugars and acetone shows quite different selectivity. For a start, cyclic acetals of acetone prefer to be five- rather than six-membered rings. In a six-membered ring, one of the acetone's methyl groups would have to be axial, so the five-membered ring is preferred. A 5,5 or 5,6 ring fusion is more stable if it is *cis*, and so acetone acetals (acetonides) form preferentially from *cis* 1,2-diols. Glucose has no neighbouring *cis* hydroxyls in the pyranose form, but in the furanose form it can have two pairs. Formation of an acetal with acetone fixes glucose in the furanose form. This is all summarized in the scheme below.



The open-chain form of glucose is in equilibrium with both the pyranose and the furanose forms through reversible hemiacetal formation using the black and green OH groups, respectively. Normally, the pyranose form is preferred, but the furanose form can form a double acetal with acetone, one acetal having two *cis*-fused five-membered rings and the other being on the side chain. This double acetal is the product isolated from the reaction.

If we want to fix glucose in the open-chain form, we must make an 'acetal' of quite a different kind using a thiol (RSH) instead of an alcohol, an aldehyde, or a ketone. The thiol combines with the aldehyde group of the open-chain form to give a stable dithioacetal. The dithioacetal is evidently more stable than the alternative hemiacetals or monothioacetals that could be formed from the pyranose or furanose forms.

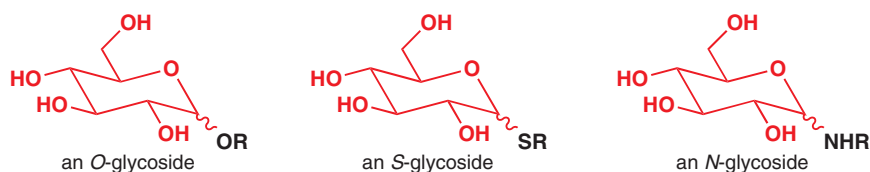


■ The most important *N*-glycosides are, of course, the nucleotides, which we have already described in some detail.

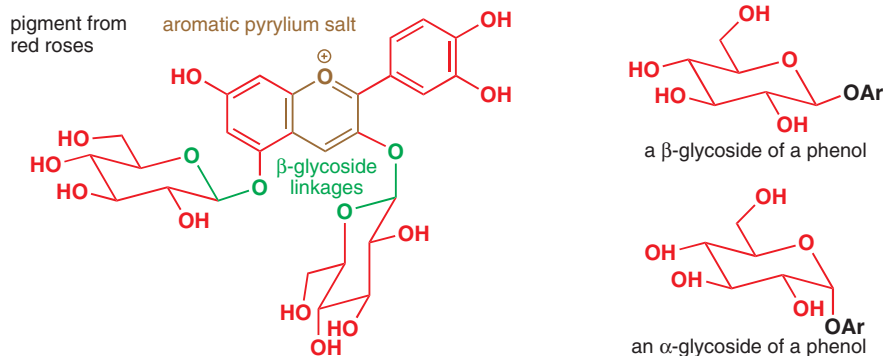
➡ You saw an example in Chapter 6 (p. 129) where acetone cyanohydrin is found in the cassava plant as a glucoside.

Glycosides in nature

Many alcohols, thiols, and amines occur in nature as glycosides, that is as *O*-, *S*-, or *N*-acetals at the anomeric position of glucose. The purpose of attaching these compounds to glucose is often to improve solubility or transport across membranes—to expel a toxin from the cell, for example. Sometimes glucose is attached in order to stabilize the compound so that glucose appears as nature's protecting group, rather as a chemist might use a THP group (Chapter 23).



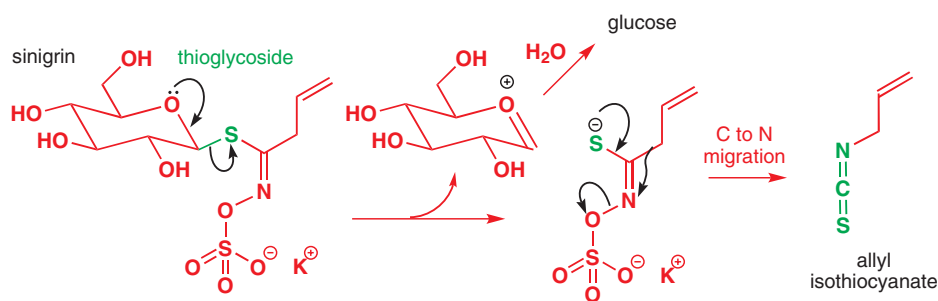
O-Glycosides occur in immense variety with glucose and other sugars being joined to the OH groups of alcohols and phenols to form acetals. The stereochemistry of these compounds is usually described by the Greek letters α and β . If the OR bond is down, it's an α -glycoside; if up, a β -glycoside. An attractive example is the pigment of red roses, which is an interesting aromatic oxygen heterocycle (an anthocyanidin). Two of the phenolic OH groups are present as β -glycosides.



α - and β -glycosides

It is easy to remember which is which, as long as you accept that people who devise nomenclature must be maliciously foolish. Just as *E* means *trans* and *Z* means *cis* (each letter has the shape of the *wrong* isomer), so α means *below* and β means *above*—each word begins with the *wrong* letter.

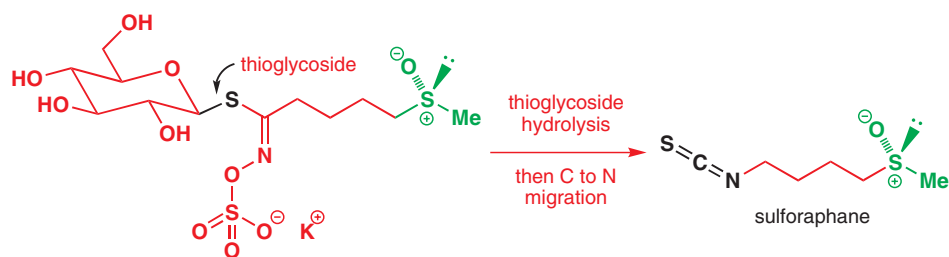
It's often proposed that there are special benefits to health in eating broccoli and brussels sprouts because of the sulfur-containing antioxidants they contain. These compounds are unstable isothiocyanates. They are not usually present in the plant; damage—by cutting or cooking, for example—induces a glycosidase (an enzyme which hydrolyses glycosides) to release the sulfur compound from its glucose protection. A simple example is sinigrin. The S-glycosides of the sinigrin group start to hydrolyse in the same way. The sulfur atom is the better leaving group when it leaves as an anion (though worse than oxygen when the hydrolysis occurs in acidic conditions) and the anion is additionally stabilized by conjugation.



The next step is surprising. A rearrangement occurs, rather similar to the Beckmann rearrangement, in which the alkyl group migrates from carbon to nitrogen and an isothiocyanate ($R-N=C=S$) is formed. Sinigrin occurs in mustard and horseradish, and it is the release of the allyl isothiocyanate that gives these their 'hot' taste. When mustard powder is mixed with water, the hot taste develops over some minutes as sinigrin is hydrolysed to the isothiocyanate.

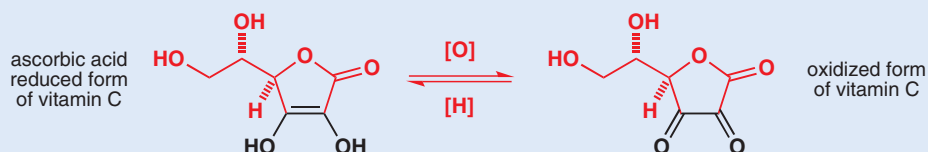
The S-glycoside in broccoli and brussels sprouts that is proposed to offer protection from cancer is somewhat similar but has one more carbon atom in the chain and contains a sulfoxide group as well. Hydrolysis of the S-glycoside is followed by the same rearrangement, producing a molecule called sulforaphane. Sulforaphane protects against cancer-causing oxidants by inducing the formation of a reductive enzyme.

➡ The Beckmann rearrangement is described in Chapter 36, p. 958.



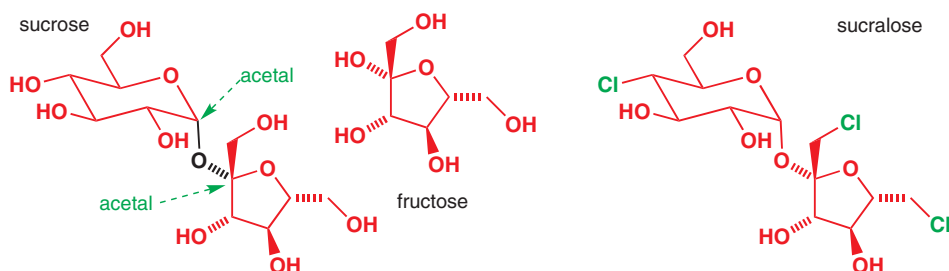
Vitamin C is a derivative of glucose

Nature makes some important compounds from simple sugars. Vitamin C—ascorbic acid—is one of these. It certainly looks very like a sugar as it has six carbon atoms, each having an oxygen atom as substituent as well as an oxygen heterocycle. Like glutathione, it protects cells from stray oxidants as well as being involved in primary redox pathways (we mentioned earlier its role in collagen synthesis). Its reduced and oxidized forms are shown below.



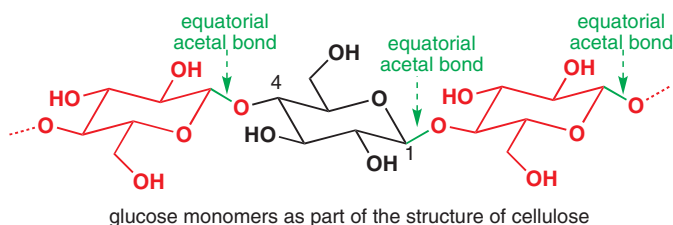
Most sugars are embedded in complex carbohydrates

The most familiar of all sugars is sucrose—the mixed acetal formed from glucose and fructose. Sucrose is of course sweet, and is easily metabolized into fats. But if three of the OH groups in sucrose are replaced by chlorine atoms, a compound 600 times as sweet is produced: less of it is needed to get the same sweet taste and the chlorines reduce the rate of metabolism so that much less fat is made. This is the compound sucralose, discovered by chemists at Tate & Lyle and now used to sweeten soft drinks.

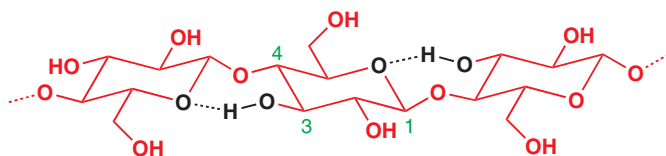


10^{15} kg per year of cellulose is literally an astronomical amount: it's about the mass of one of the moons of Mars, Deimos. Our moon weighs 10^{22} kg.

Sucrose is a disaccharide—two simple sugars linked by an acetal. In general, saccharides have the same relationship to sugars as peptides and proteins have to amino acids. One of the most abundant compounds in nature is a saccharide: cellulose, the structural material of plants. It is a glucose polymer and is produced in simply enormous quantities (about 10^{15} kg per year). Each glucose molecule is joined to the next through an acetal formed by attack of the C4 hydroxyl group of one glucose molecule on the anomeric carbon atom of the next. Here is that basic arrangement.



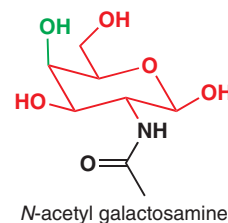
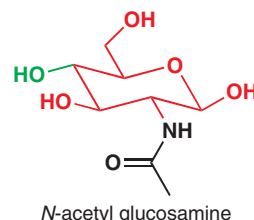
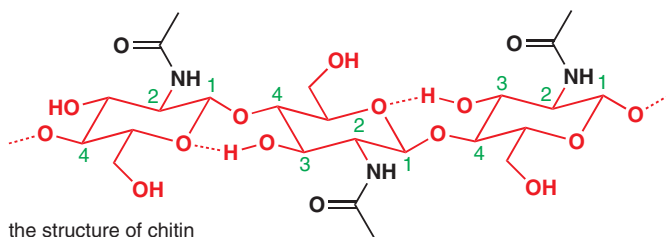
Notice that the anomeric bonds are all equatorial. This means that the cellulose molecule is linear in general outline. It is made rigid by extra hydrogen bonds between the 3-OH groups and the ring oxygen atoms—like this.



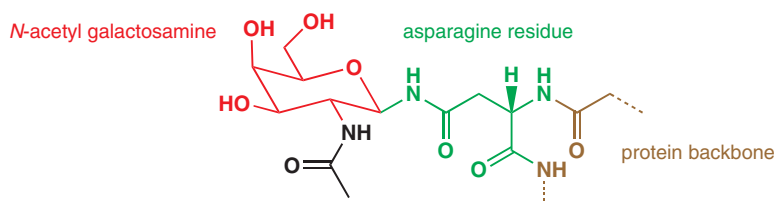
The polymer is also coiled to increase stability still further. All this makes cellulose very difficult to hydrolyse, and humans cannot digest cellulose as we do not have the necessary enzymes. Other mammals have evolved devices such as multiple stomachs (in ruminants, such as cattle) to enable them to degrade cellulose.

Amino sugars add versatility to saccharides

Amino sugars are carbohydrates into which nitrogen is incorporated. These molecules allow proteins and sugars to combine and produce structures of remarkable variety and beauty. The most common amino sugars are *N*-acetyl glucosamine and *N*-acetyl galactosamine, which differ only in stereochemistry. The hard outer skeletons of insects and crustaceans contain chitin, a polymer very like cellulose but made of acetyl glucosamine instead of glucose itself. It coils up in a similar way and provides the toughness of crab shells and beetle cases.

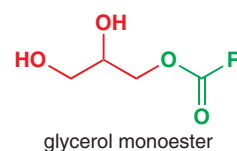
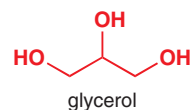


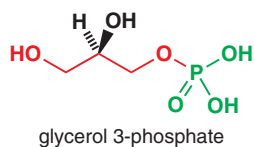
Cell membranes must not be so impermeable as they need to allow the passage of water and complex molecules. These membranes contain *glycoproteins*—proteins with amino sugar residues attached to asparagine, serine, or threonine in the protein. The attachment is at the anomeric position so that these compounds are *O*- or *N*-glycosides of the amino sugars. The structure below shows *N*-acetyl galactosamine attached to an asparagine residue as an *N*-glycoside.



Lipids

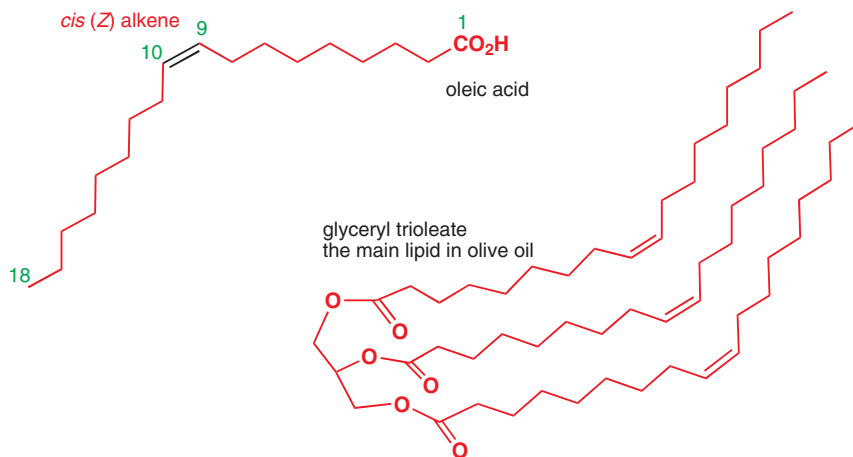
Lipids (fats) are the principal components of cell membranes. Along with cholesterol, also a component of the cell membrane, they have acquired a bad name, but they are nonetheless essential to the function of membranes as selective barriers to the movement of molecules. The most common types of lipids are esters of glycerol. Glycerol is just propane-1,2,3-triol but it has interesting stereochemistry. It is not chiral as it has a plane of symmetry, but the two primary OH groups are enantiotopic. If one of them is modified—by esterification, for





example—the molecule becomes chiral. Natural glycerol 3-phosphate is such an ester and it is optically active.

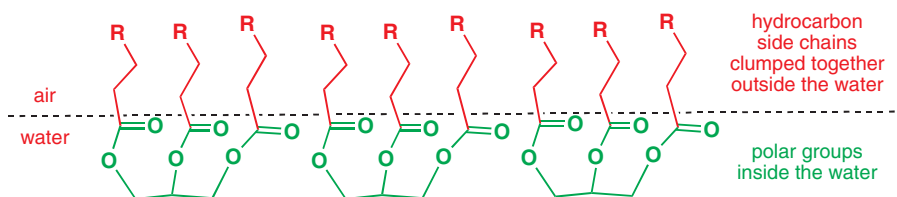
A typical lipid in foodstuffs is the triester formed from glycerol and oleic acid, which is the most abundant lipid in olive oil. Oleic acid is a mono-unsaturated fatty acid—it has one *Z* double bond in the middle of the C_{18} chain. This bond gives the molecule a marked kink in the middle. The compound actually present in olive oil is the triester, also kinked.



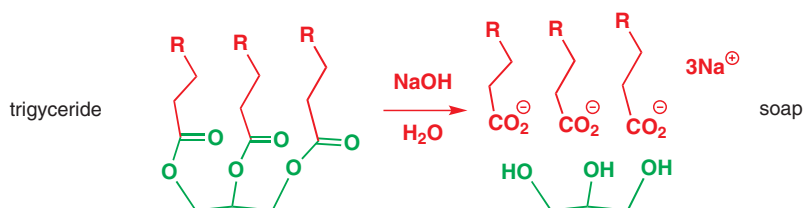
Oil and water do not mix

■ You may have done the Langmuir trough experiment in a physical chemistry practical class. This involves measuring the size of a molecule by allowing an oil to spread on the surface of water in a unimolecular layer.

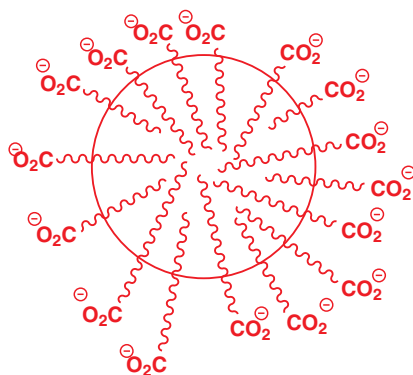
The lipid has, more or less, the conformation shown in the diagram with all the polar ester groups at one end and the hydrocarbon chains bunched together in a non-polar region. Oil and water do not mix, it is said, but triglyceride lipids associate with water in a special way. A drop of oil spreads out on water in a very thin layer. It does so because the ester groups sit inside the water and the hydrocarbon side chains stick out of the water and associate with each other.



When triglycerides are boiled with alkali, the esters are hydrolysed and a mixture of carboxylate salts and glycerol is formed. This is how soap is made—hard soap is the sodium salt and soft soap the potassium salt.



When a soap is suspended in water, the carboxylate groups have a strong affinity for the water and so oily globules or **micelles** are formed with the hydrocarbon side chain inside. It is these globules that remove greasy dirt from you or your clothes.



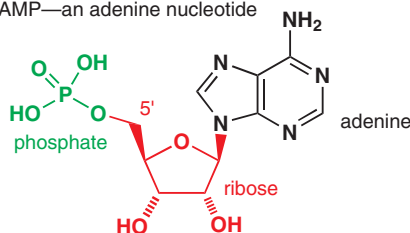
Mechanisms in biological chemistry

Nature uses the same chemistry as we do in the laboratory, and to do that chemistry she needs reagents. Chemists are free to use temperatures typically ranging 100 °C either side of 20 °C, any solvents they choose, inert or reactive atmospheres, and so on. Not so nature: all nature's reagents must work at ambient temperature in the presence of water and in the presence of a reactive gas, oxygen. In this section we will survey some of nature's solutions to these challenges.

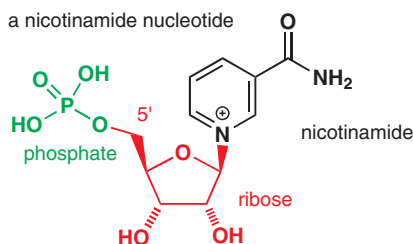
Nature's NaBH_4 is a nucleotide: NADH or NADPH

You met nucleotides, and their role in the structure of nucleic acids, earlier in this chapter. Nature also uses nucleotides as reagents. Here is the structure of AMP, just to remind you of a structure you met before, side by side with a pyridine-containing nucleotide.

AMP—an adenine nucleotide

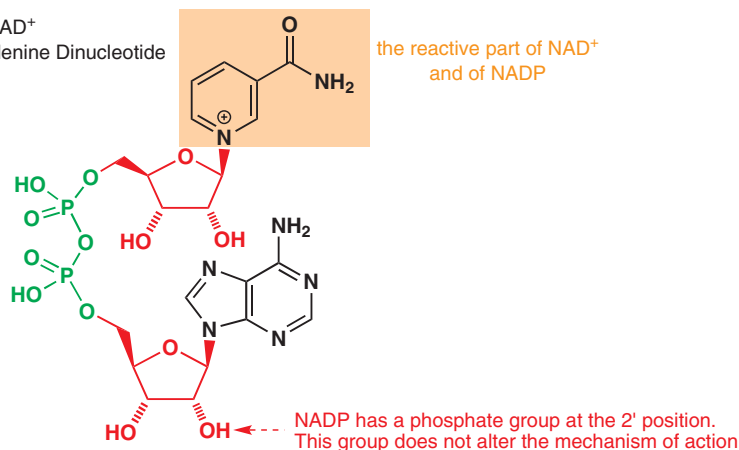


a nicotinamide nucleotide

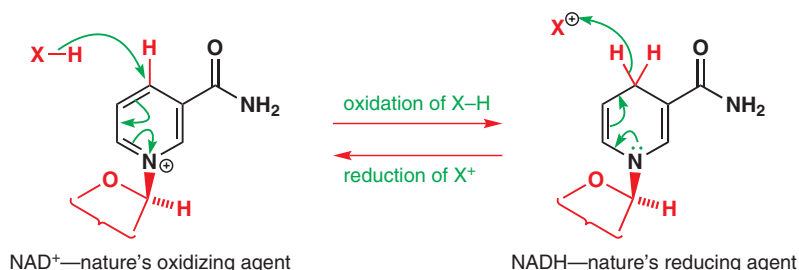


These two nucleotides can combine together as a pyrophosphate to give a dinucleotide called nicotinamide adenine dinucleotide, or NAD (or NAD^+ —note the positively charged pyridinium). Notice that the link is not at all the same as in the nucleic acids. The latter are joined by one phosphate that links the 3' and 5' positions. Here we have a pyrophosphate link between the two 5' positions.

NAD^+
Nicotinamide Adenine Dinucleotide

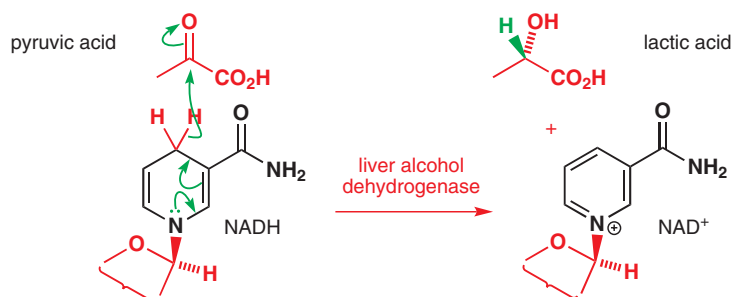


The positively charged pyridinium ring is the part of the molecule which does all the work and from now on we will draw only the reactive part for clarity. NAD^+ is one of nature's most important oxidizing agents. Some biochemical pathway reactions use NADP instead, but this differs only in having an extra phosphate group on the adenosine portion so the same part structure will do for both. NAD^+ and NADP both work by accepting a hydrogen atom and a pair of electrons from another compound. The reduced compounds are called NADH and NADPH.

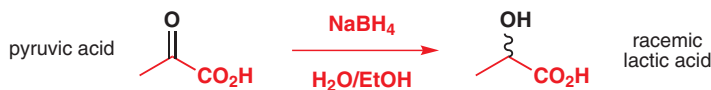


The names of enzymes are usually chosen to tell us where they come from and what job they do and the name ends '-ase'. A **dehydrogenase** is a redox enzyme which catalyses the removal (or, as in this case, the addition) of hydrogen.

The reduction of NAD^+ (and NADP) is reversible, and NADH is itself a reducing agent. We will first look at one of its reactions: a typical reduction of a ketone. The ketone is pyruvic acid and the reduction product is lactic acid—both important metabolites. The reaction is catalysed by an alcohol dehydrogenase.

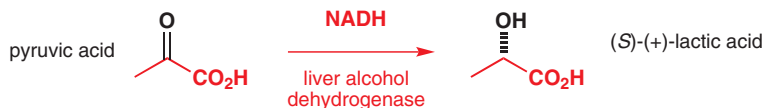


This is a reaction that would also work in the laboratory with NaBH_4 as the reducing agent, but there is a big difference. The product from the NaBH_4 reaction *must* be racemic—the starting material, reagent, and solvent are all achiral.



➡ If you are not clear about enantioselective reactions and why NaBH_4 must give a racemic mixture, re-read Chapter 41. If you are not clear about the terms 'enantiotopic' and 'prochiral' re-read Chapters 31 and 33. If you are not clear about what enantiomers are, you must re-read Chapter 14 now.

But the product from the enzymatic reaction is optically active. The two faces of pyruvic acid's carbonyl group are enantiotopic and, by controlling the addition so that it occurs from one face only, the enzyme-catalysed reaction gives a single enantiomer of lactic acid.

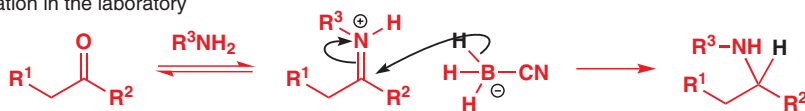


Reductive amination in nature

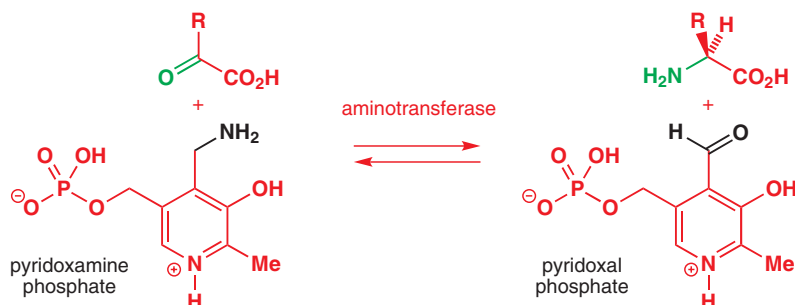
One of the best methods for making amines in the laboratory is **reductive amination**, in which an imine (formed from a carbonyl compound and an amine) is reduced to a saturated amine. Common reducing agents include NaCNBH_3 and hydrogen with a catalyst.

➡ For more on reductive amination, see Chapter 11.

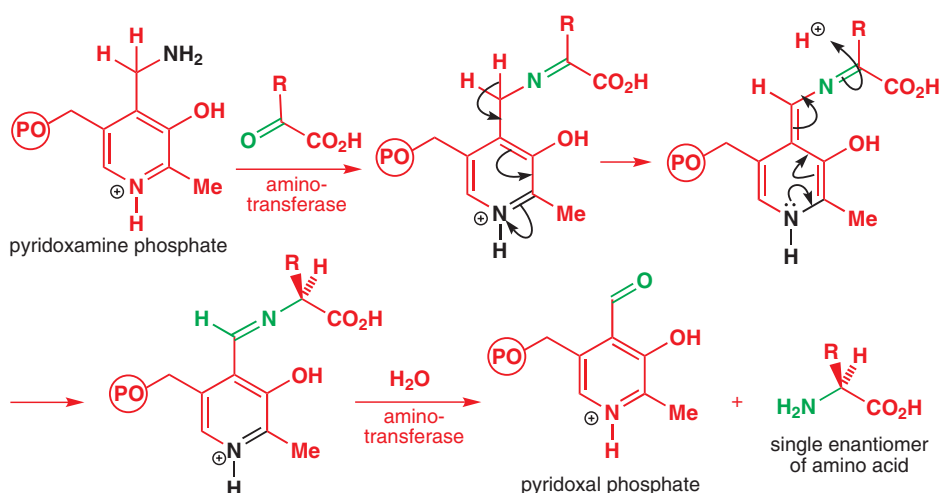
reductive amination in the laboratory



This reaction, of course, produces racemic amines. But nature transforms this simple reaction into an enantioselective and reversible one that is beautiful in its simplicity. The reagents are a pair of substituted pyridines called pyridoxamine and pyridoxal, and the enzyme is an aminotransferase.



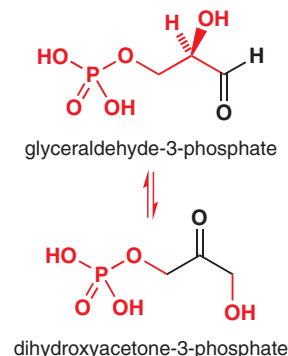
The mechanism of the amination starts with the formation of an imine from the amino group and the green carbonyl. Removal of the now very acidic proton between the protonated pyridine and the conjugated imino-carboxylic acid gives a dihydropyridine, which rearomatizes by protonation next to the carboxylic acid. This step is enantioselective, with the proton being delivered from the enzyme. Finally, hydrolysis of the new imine gives pyridoxal and a single enantiomer of the amino acid.

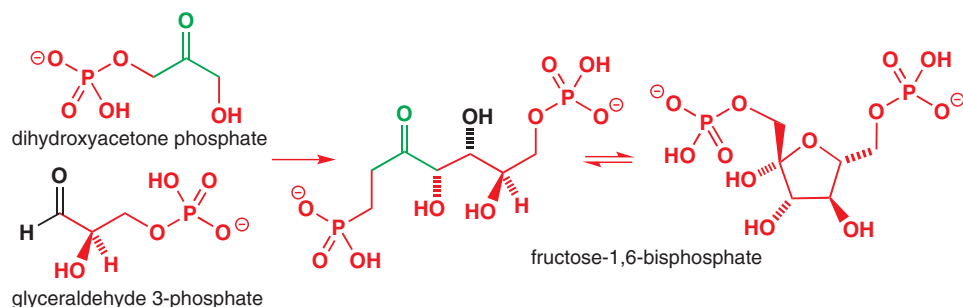


■ 'OP' or 'P' in a circle is commonly used to represent a phosphate group.

Nature's enolate equivalents: lysine enamines and coenzyme A

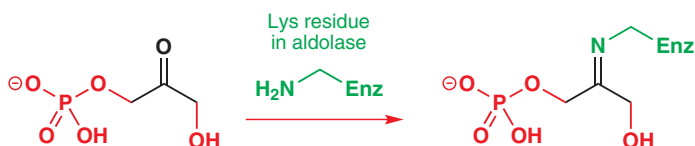
Nature breaks down glucose to produce energy, and in doing so produces smaller molecules which enter the citric acid cycle and are converted ultimately to carbon dioxide. In the other direction, the six-carbon sugar fructose can be made from two two three-carbon fragments. The key reaction in both cases is the step in which the C–C bond linking the two C_3 sugars is formed or broken. The C_3 sugars are glyceraldehyde and dihydroxyacetone, both as their phosphate esters, and the reaction between them is an aldol condensation. The enol of dihydroxyacetone phosphate attacks the electrophilic aldehyde carbonyl group of glyceraldehyde 3-phosphate, catalysed by an enzyme named aldolase. The product is a ketohexose (i.e. a six-carbon sugar with a ketone carbonyl group), fructose-1,6-bisphosphate.



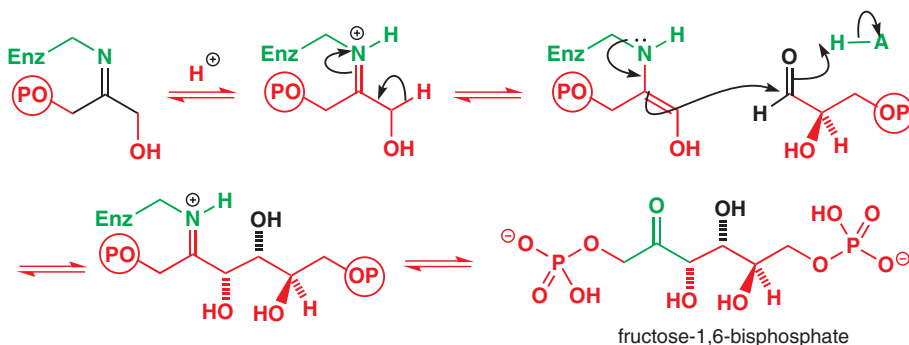


No enolate ion is formed in this aldol reaction. Instead a lysine residue in the aldolase enzyme forms an imine with the keto-triose.

■ The rest of the aldolase molecule is represented by 'Enz'.

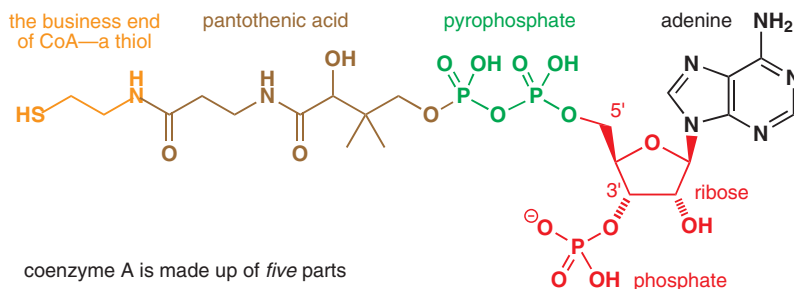


Proton transfers allow this imine to be converted into an enamine, which acts as the nucleophile in the aldol reaction. Stereochemical control (it's a *syn* aldol) comes from the way in which the two molecules are held by the enzyme as they combine. The product is the imine, which is hydrolysed to the open-chain form of fructose-1,6-bisphosphate.



Many other reactions in nature use enamines, mostly those formed from lysine. However, a more common enol equivalent is based on thiol esters derived from coenzyme A. Coenzyme A is an adenine nucleotide at one end, linked by a 5'-pyrophosphate to pantothenic acid, a compound that looks rather like a tripeptide, and then to an amino thiol. Here is the structure broken down into its parts.

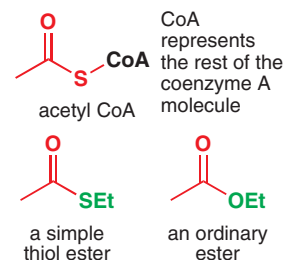
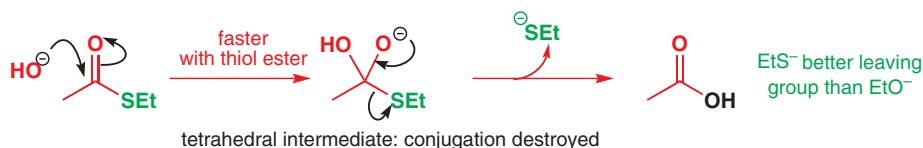
■ Compare this structure with that of NAD—the adenine nucleotide is the same, as is the 5'-pyrophosphate link. The difference is at the other end of that link where we find this new tripeptide-like molecule and not another nucleotide. There is also a 3'-phosphate on the ribose ring not present in NAD (and note that while NADP contains a phosphorylated ribose, its phosphate is on the 2' hydroxyl group!).



By now you will realize that most of this molecule is there to allow interaction with the various enzymes that catalyse the reactions of coenzyme A. Coenzyme A is conveniently abbreviated in structures to CoASH, where the SH is the vital thiol functional group, and all the

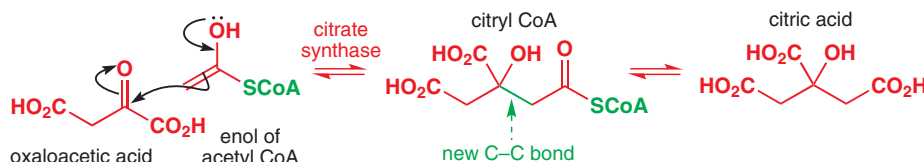
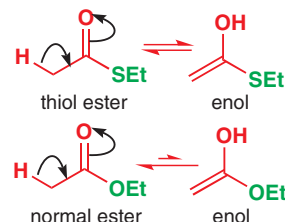
reactions we will be interested in are those of esters of CoASH. These are thiol esters, as opposed to normal alcohol esters, and the difference is worth a few comments.

Thiol esters are less conjugated than ordinary esters, and ester hydrolysis occurs more rapidly with thiol esters than with ordinary esters because in the rate-determining step (nucleophilic attack on the carbonyl group) there is less conjugation to destroy. The thiolate is also a better leaving group.



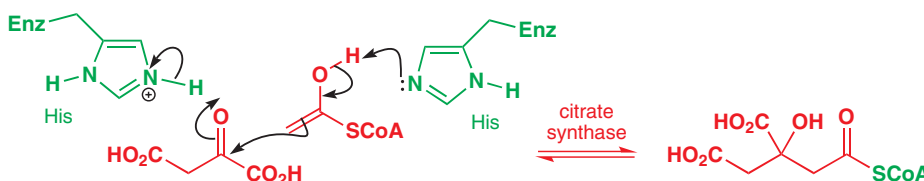
Another reaction that goes better with thiol esters than with ordinary esters is enolization. This is an equilibrium reaction and the enol has lost the conjugation present in the ester. Again, a thiol ester has less to lose so is more enolized, and it is the enolization of thioesters of coenzyme A that we are now going to discuss.

We mentioned the citric acid cycle earlier but we have not so far discussed the chemistry involved. The citric acid cycle allows metabolism to shunt carbon atoms between small molecules, and the key step is the synthesis of citric acid from oxaloacetate and acetyl CoA. The reaction is essentially an aldol reaction between the enol of an acetate ester and an electrophilic ketone, and the enzyme which catalyses the reaction is known as citrate synthase.



The mechanism shows the enol of acetyl CoA attacking the reactive ketone. In nature all these reactions are catalysed by the enzyme. In the C-C bond-forming step, one histidine residue removes the enol proton and another histidine, in its protonated form, is placed to donate a proton to the oxygen atom of the ketone. You should see now why histidine is so useful to enzymes: its imidazole ring means it can act either as an acid or as a base at neutral pH.

➡ The pK_a of protonated imidazole is about 7: see Chapter 8.



➡ This is general acid catalysis, as described in Chapter 39.

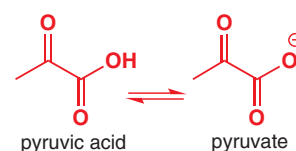
Even the hydrolysis of the reactive thiol ester is catalysed by the enzyme and histidine again functions as a proton donor, with the hydrolysis, like the enolization, being enhanced by the thiol ester.

The two enol equivalents that we have met so far are quite general: lysine enamines can be used for any aldehyde or ketone and CoA thiol esters for any ester. Another class of enol equivalent—the enol ester—has just one representative but it is a most important one.

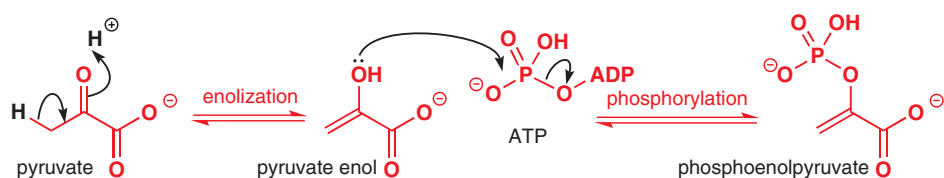
Phosphoenolpyruvate

Pyruvic acid is an important metabolite in its own right, as we shall see shortly. It is the simplest α -keto-acid (2-oxopropanoic acid). Having the two carbonyl groups adjacent makes them more reactive: the ketone is more electrophilic and enolizes more readily, and the acid is stronger.

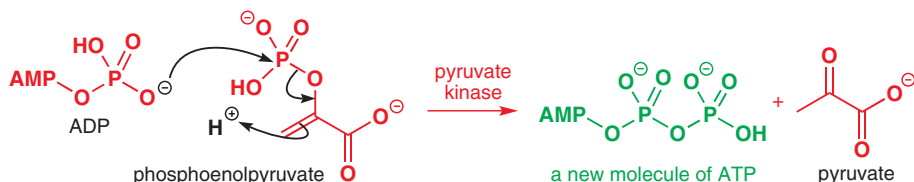
Nature uses the enol phosphate of pyruvic acid (phosphoenolpyruvate or PEP) as an important reagent. We might imagine making this compound by first forming the enol and then esterifying on oxygen by some phosphorylating agent such as ATP.



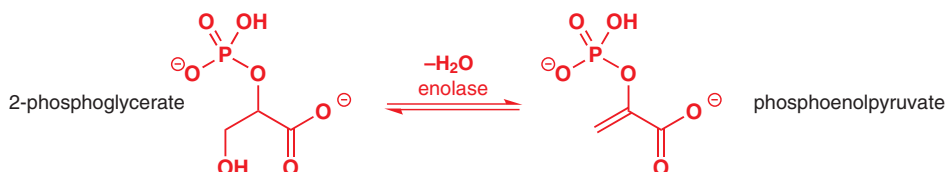
➡ For an explanation of the effect of two adjacent carbonyl groups, see Chapter 26, p. 643.



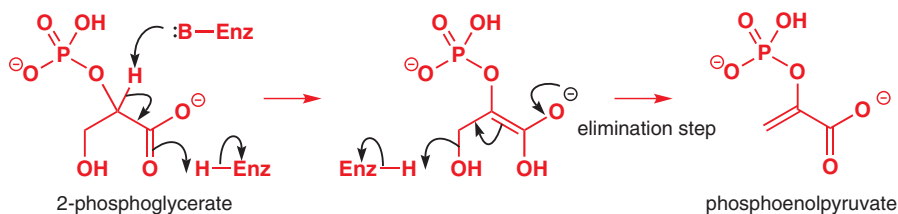
Now, in fact, this reaction does occur in nature as part of the glycolysis pathway, but it occurs almost entirely in reverse. PEP is used as a way to make ATP from ADP during the oxidation of energy-storing sugars. An enol is a better leaving group than an ordinary alcohol, especially if it can be protonated at carbon. The reverse reaction might look like this.



PEP is also used as an enol in the making of carbon-carbon bonds when the electrophile is a sugar molecule. But if PEP is not made by enolization of pyruvate, how is it made? The answer is by dehydration. The phosphate is already in place when the dehydration occurs, catalysed by the enzyme enolase.



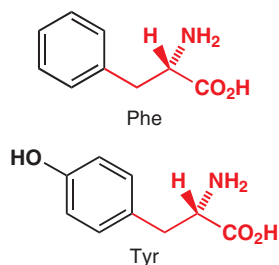
You saw in Chapter 17 how simple OH groups can be lost in dehydration reactions. Either the OH group was protonated by strong acid (this is not an option in living things) or an enol or enolate pushed the OH group out in an E1cB-like mechanism. This must be the case here as the better leaving group (phosphate) is ignored and the worse leaving group (OH) expelled.

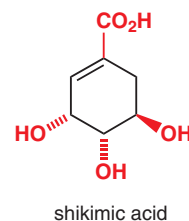
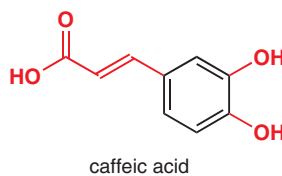
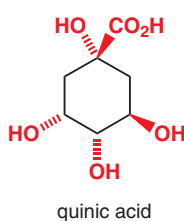
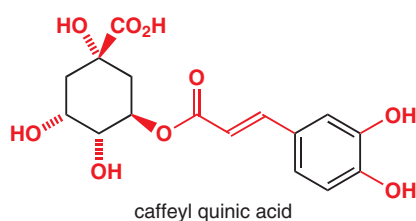


The shikimic acid pathway

The shikimic acid pathway is responsible for the biosynthesis of a large number of aromatic compounds, particularly in plants. Most important for many mammals is the fact that plants manufacture the aromatic amino acids Phe (phenylalanine), Tyr (tyrosine), and Trp (tryptophan). These are 'essential' amino acids for humans—we have to have them in our diet as we cannot make them ourselves.

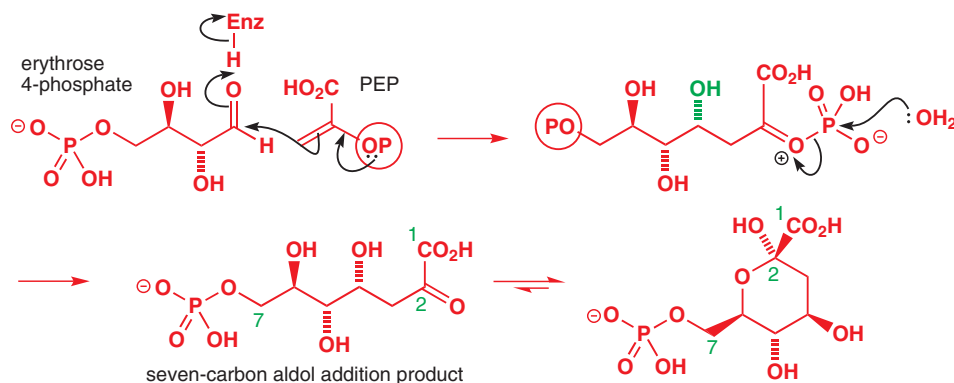
So how do plants make aromatic rings? A clue to the chemistry involved comes from the structure of caffeoyl quinic acid, a compound that forms about 13% of the soluble solids from coffee beans. A substantial proportion of instant coffee is caffeoyl quinic acid.



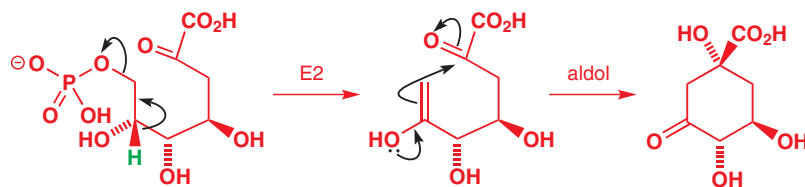
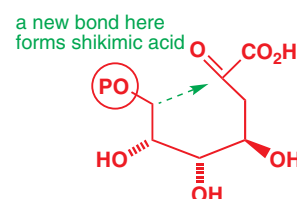


This ester has two six-membered rings—one aromatic and one saturated. You might imagine making an aromatic ring by the dehydration (losing three molecules of water) of a cyclohexane triol and the saturated ring in caffeoyl quinic acid looks a good candidate. It is now known that both rings (shown in black) come from the same intermediate, shikimic acid.

This key intermediate has given its name to nature's general route to aromatic compounds and many other related six-membered ring compounds: the shikimic acid pathway. This pathway contains some of the most interesting reactions (from a chemist's point of view) in biology. It starts with an aldol reaction between phosphoenolpyruvate as the nucleophilic enol component and the C₄ sugar erythrose 4-phosphate as the electrophilic aldehyde.

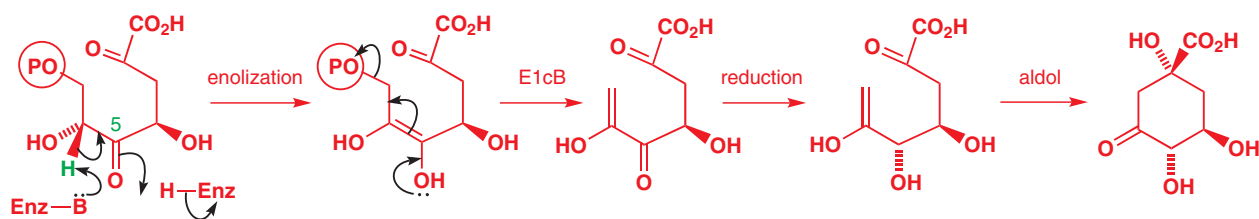


Hydrolysis of the phosphate releases the aldol product, a C₇ α-keto-acid with one new stereogenic centre, which is in equilibrium with a hemiacetal, just like a sugar. This intermediate has the right number of carbon atoms for shikimic acid and the next stage is a cyclization. If we redraw the uncyclized C₇ α-keto-acid in the right shape for cyclization we can see what is needed. The green arrow shows which bond needs to be formed. This bond could be formed by another aldol reaction, and there is an obvious route to the required enol by elimination of phosphate. However, this would require the removal of a proton (green in the diagram) that is not at all acidic.



The problem can be avoided if the hydroxyl group at C₅ is first oxidized to a ketone (using NAD⁺ as the oxidant). Then the green proton is much more acidic, and the elimination becomes an E1cB reaction, similar to the one in the synthesis of PEP. True, the ketone must be reduced back to the alcohol afterwards but nature can deal with that easily. There are obviously several more steps to get to shikimic acid but all the C–C bonds are in place, the most significant of them being formed by aldol reactions.

Web link: You can find more on the shikimic acid pathway in the online chapter 'Mechanisms in biological chemistry'.



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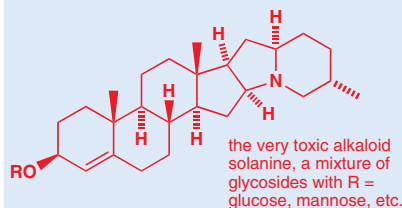
Natural products

Organic chemists mean something particular by the phrase 'natural products'. Of course, all the compounds we have so far discussed are natural and their chemistry is common to most living things. But living things also make chemicals by the processes of secondary metabolism that are found in few, if any, other organisms. The flavouring principles of herbs and fruit, the antibiotics from moulds and the toxic alkaloids in plants are all examples. These compounds are what we mean by 'natural products', especially if they are useful to humans.

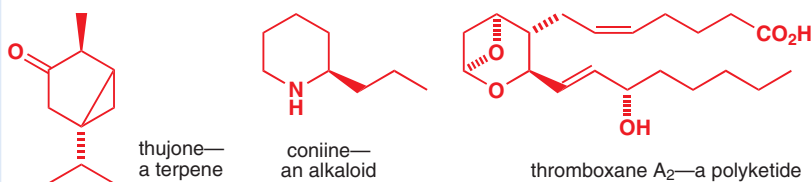
Natural products often seem to have little value to the organism itself, and are made by the processes of secondary metabolism. They are classified by the way they are made into terpenes and steroids, alkaloids, and polyketides.

Solanaceae alkaloids

The Solanaceae family includes not only deadly nightshade (*Atropa belladonna*—hence atropine) plants but also potatoes and tomatoes. Parts of these plants also contain toxic alkaloids, for example you should not eat green potatoes because they contain the toxic alkaloid solanine.



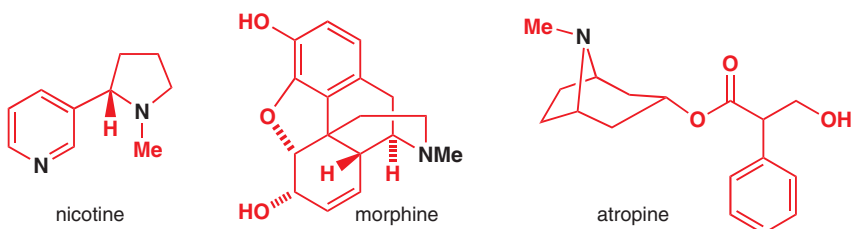
Atropine is a racemic compound but the (*S*)-enantiomer occurs in henbane (*Hyoscyamus niger*) and was given a different name, hyoscyamine, before the structures were known. In fact, hyoscyamine racemizes very easily just on heating in water or on treatment with weak base. This is probably what happens in the deadly nightshade plant.



Thujone is a terpene that is thought to be the poisonous principle in absinthe—the drink that reduced many artists and writers to idiocy in Paris around 1900. Coniine is an alkaloid and the poison in hemlock with which Socrates was executed. Thromboxane is a polyketide involved in blood-clot formation and is a human natural product.

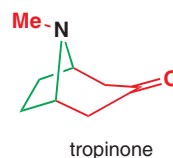
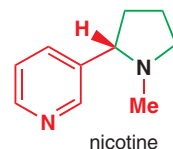
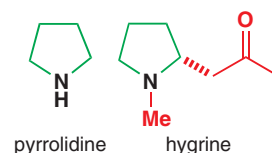
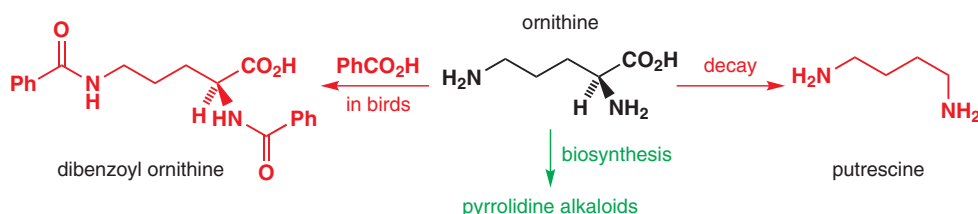
Alkaloids are made by amino acid metabolism

Alkaloids were known in ancient times because they are easy to extract from plants and some of them have powerful and deadly effects. Any plant contains thousands of chemical compounds, but some plants, like the deadly nightshade, can be mashed up and extracted with aqueous acid to give a few compounds soluble in that medium, which precipitate on neutralization. These compounds were seen to be 'like alkali' and in 1819 Meissner, the apothecary from Halle, named them 'alkaloids'. Lucrezia Borgia already knew all about this and put the deadly nightshade extract atropine in her eyes (to make her look beautiful: atropine dilates the pupils) and in the drinks of her political adversaries to avoid any trouble in the future. Now, we would simply say that they are basic because they are amines. Below is a selection with the basic amino groups marked in black. Natural products are often named by a combination of the name of the organism from which they are isolated and a chemical part name. These compounds are all *amines* so all their names end in '-ine'. They appear very diverse in structure but all are made in nature from amino acids.

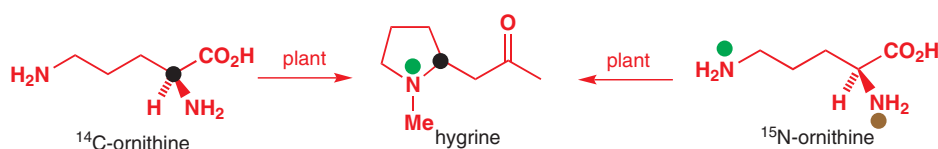


Pyrrolidine alkaloids are made from the amino acid ornithine

Pyrrolidine is the simple five-membered cyclic amine and pyrrolidine alkaloids such as nicotine contain this ring. All are made in nature from ornithine. Ornithine is an amino acid not usually found in proteins (it's one carbon atom shorter than lysine) but most organisms use it, often in the excretion of toxic substances. If birds are fed benzoic acid (PhCO_2H) they excrete dibenzoyl ornithine. When dead animals decay, the decarboxylation of ornithine leads to putrescine, the smell of rotten meat.

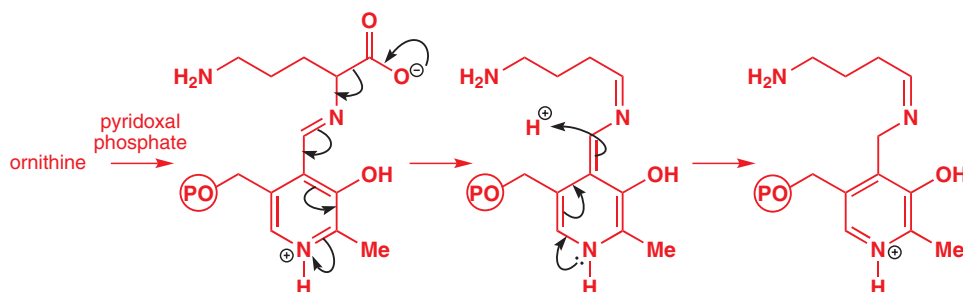


Biosynthetic pathways are usually worked out by isotopic labelling of potential precursors and in the schemes below the isotopically labelled atom is shown with a coloured blob. Some plants—notably the coca plant—produce the simple pyrrolidine alkaloid hygrine, which we will take as an illustration. If ornithine is made with a ^{14}C label at its α position and fed to the plant, labelled hygrine is isolated. If each amino group in ornithine is labelled in turn with ^{15}N , the α amino group is lost but the γ amino group is retained.



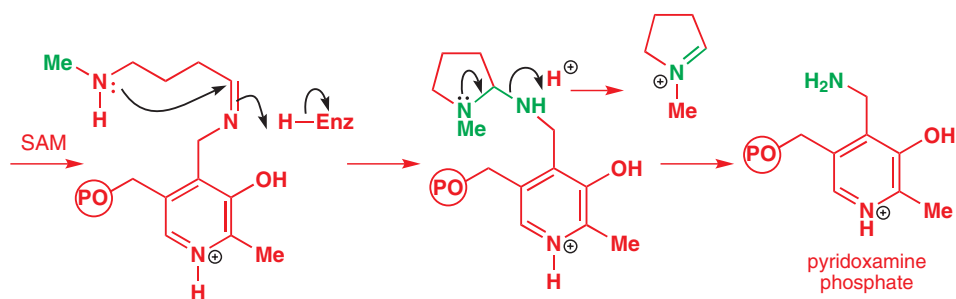
Further labelling experiments along these lines showed that the CO_2H group as well as the α amino group was lost from ornithine and that the rest of the molecule makes the pyrrolidine ring. The three-carbon side chain in hygrine comes from acetate, or rather from acetyl CoA, and the *N*-methyl group comes from (*S*)-adenosyl methionine (SAM, see p. 1136).

Labelling studies such as these tell us the origin of the atoms in the natural product, and we can now work through the biosynthesis—how the molecule is put together from those precursors. The first step is a pyridoxal-catalysed decarboxylation of ornithine.

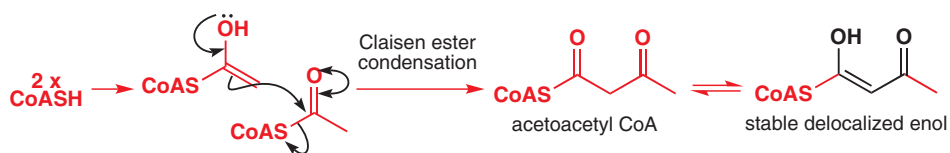


➡ You saw pyridoxal phosphate becoming involved in a reductive amination on p. 1151: here—and in other biochemical pathways too—a similar mechanism leads to decarboxylation.

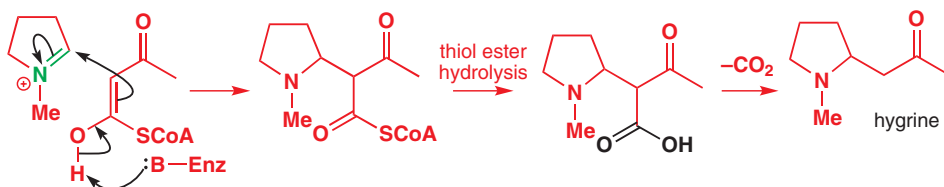
Now the terminal amino group is methylated by SAM and the secondary amine cyclizes onto the pyridoxal imine to give an aminor. Decomposition of the aminor the other way round expels pyridoxamine and releases the salt of an electrophilic imine.



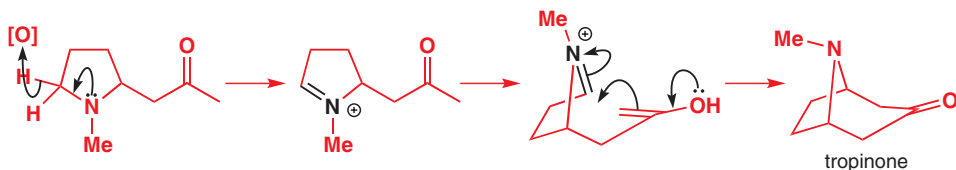
The rest of the hygrine structure comes from two molecules of acetyl CoA. We saw earlier in this chapter that the thiol ester is a good electrophile and also enolizes easily. We need both reactivities now in a Claisen ester condensation of acetyl CoA. The new keto-ester is very like the acetoacetates we used in Chapter 25 to make stable enolates and the CoA thiol ester will exist mainly as its enol, stabilized by conjugation.



The cell has a good stock of acetyl CoA and its condensation product, and as soon as the iminium ion above is generated, it is attacked by the acetoacetyl CoA. All that remains to form hygrine is the hydrolysis of the CoA thiol ester and decarboxylation of the keto-acid. This is standard chemistry, but you should ensure that you can draw the mechanisms for these steps.

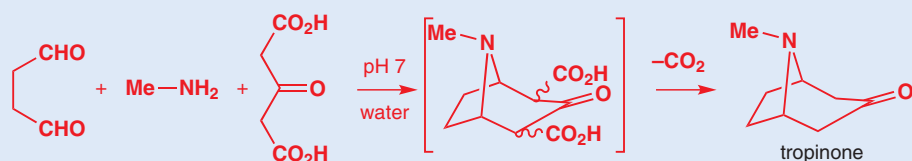


Tropinone is made from hygrine and it is clear what is needed. The methyl ketone must enolize and it must attack another iminium ion resembling the first but on the other side of the ring. A biological oxidant such as NADP is needed.



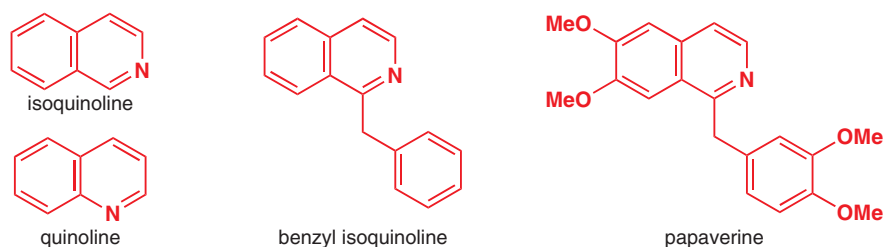
Robinson's tropinone synthesis

This complex route to tropinone was imitated as long ago as 1917 in one of the most celebrated reactions of all time, Robinson's tropinone synthesis. Robinson argued on purely chemical grounds that the sequence of imine salts and enols, which later (as shown in 1970) turned out to be nature's route, could be produced under 'natural' conditions (aqueous solution at pH 7) from a C₄ dialdehyde, MeNH₂, and acetone dicarboxylic acid. It worked and the intermediates must be very similar to those in the biosynthesis.

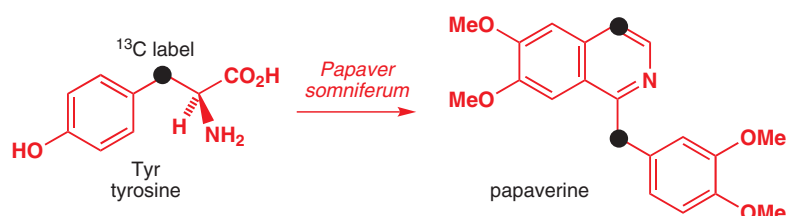


Benzyl isoquinoline alkaloids are made from tyrosine

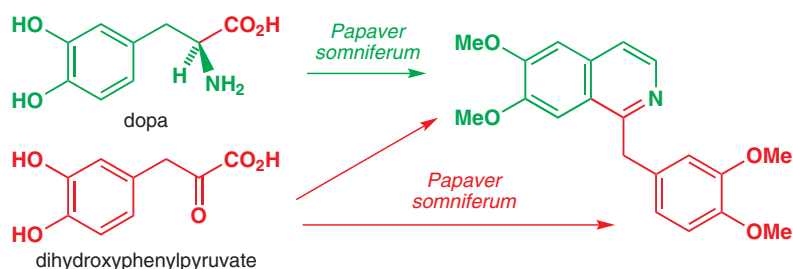
The benzyl isoquinolines are another family of alkaloids of rather different structure. They all have a benzyl group attached to position 2 of an isoquinoline ring. Usually the alkaloids are oxygenated on the benzene ring and many are found in opium poppies (*Papaver somniferum*). For all these reasons papaverine is an ideal example.



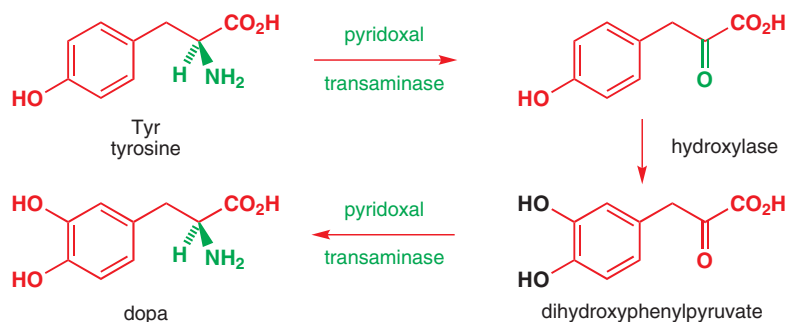
Labelling shows that these alkaloids come from two molecules of tyrosine. One must lose CO_2 and the other NH_3 . We can easily see how to divide the molecule in half, but the details will have to wait a moment.



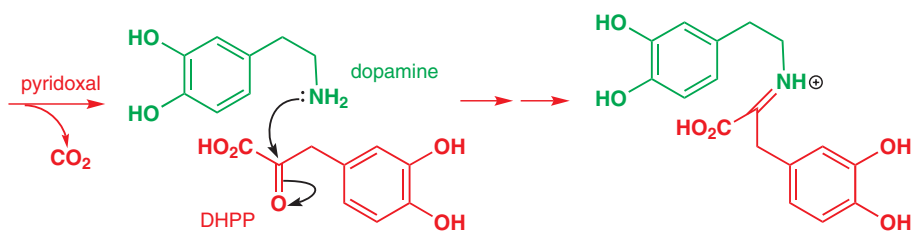
The question of when the extra OH groups are added was also solved by labelling and it was found that dihydroxyphenyl pyruvate (DHPP) was incorporated into both halves but the dihydroxyphenylalanine (an important metabolite, and also a useful medicine, usually called dopa) was incorporated only into the isoquinoline half.



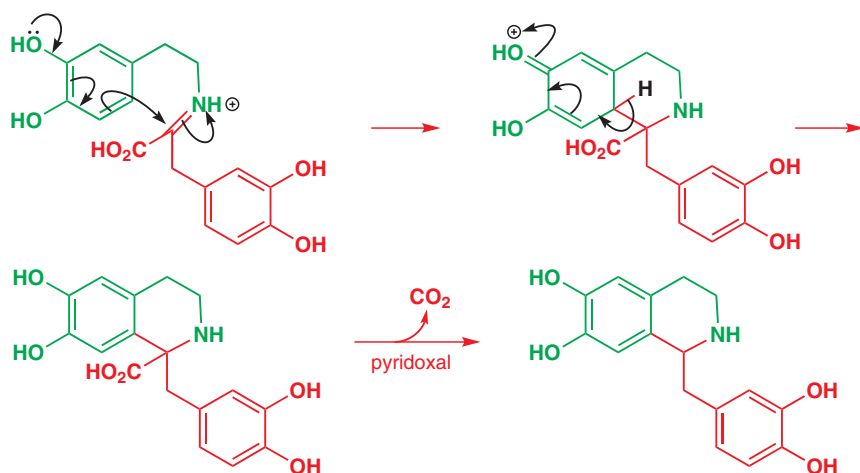
The amino acid and the keto-acid are related by a pyridoxal-mediated transaminase and the hydroxylation must occur right at the start.



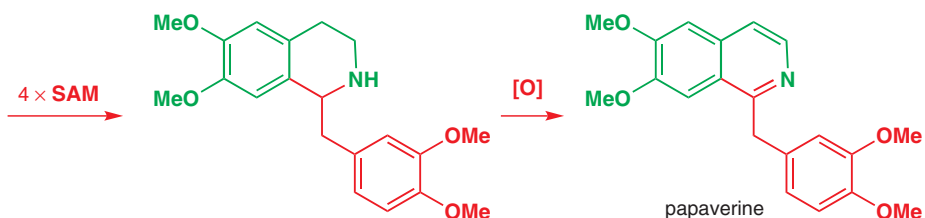
Pyridoxal-mediated decarboxylation of dopa gives dopamine and this reacts with the keto-acid to form an iminium ion perfectly placed for an intramolecular electrophilic aromatic substitution by the electron-rich dihydroxyphenyl ring.



This closes the isoquinoline ring in a Mannich-like process with the phenol replacing the enol in the pyrrolidine alkaloid biosynthesis.

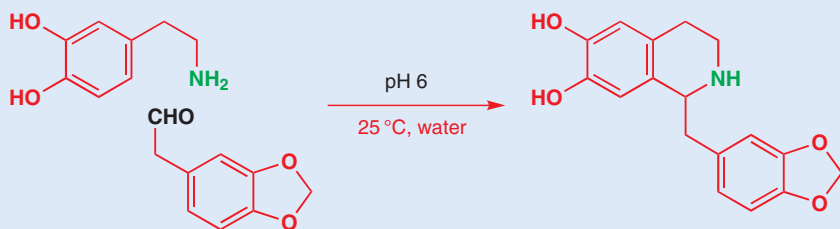


The cyclization product is still an amino acid and it can be decarboxylated by pyridoxal. Now we have something quite like papaverine but it lacks the methyl groups and the aromatic heterocyclic ring, which are introduced by methylation with SAM and oxidation.

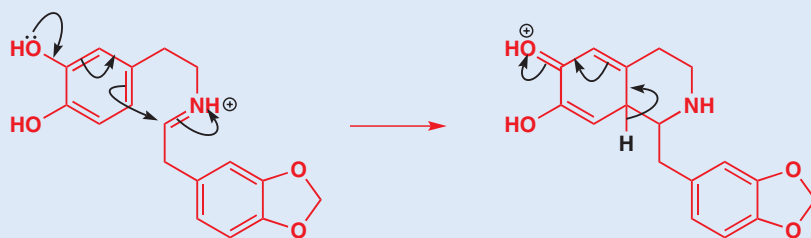


Synthesis of isoquinolines

As with tropinone, it is possible to make benzyl isoquinoline alkaloids very simply under mild conditions in the laboratory, providing that we use an aldehyde as the carbonyl component. The reaction (sometimes known as the Pictet–Spengler reaction) gives a reduced heterocyclic ring, as does the biosynthesis, but chemical oxidation can be used to give the isoquinoline.



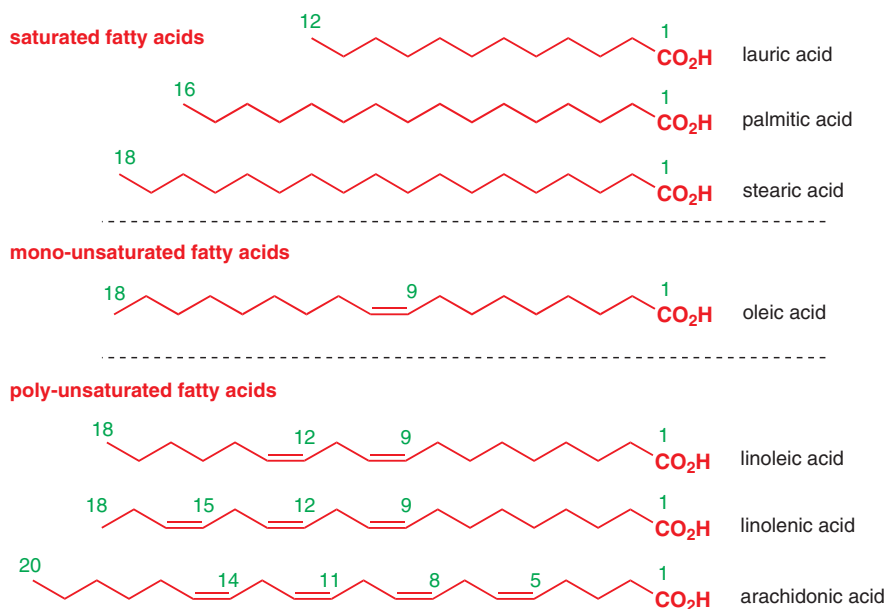
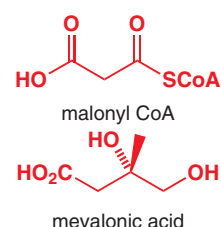
The mechanism is straightforward—the imine is formed and will be protonated at pH 6, ready for the C–C bond formation, which is both a Mannich reaction and an electrophilic aromatic substitution.



Notice that it was not necessary to protect the OH groups—the acetal on the lower ring is not for protection, and this group (methylenedioxy or dioxolane) is present in many benzyl isoquinoline alkaloids. It is formed in nature by oxidation of an MeO group *ortho* to an OH group on a benzene ring.

Fatty acids and other polyketides are made from acetyl CoA

In the last part of this chapter we will show how nature can take a very simple molecule—acetyl CoA—and build it up into an amazing variety of structures. There are two main pathways from acetyl CoA through malonyl CoA and mevalonic acid and each gives rise to two important series of natural products. Malonyl CoA leads to fatty acids and polyketides while mevalonic acid gives terpenes and steroids. We start with the simplest, the fatty acids. The list below shows just a few of the fatty acids that exist: all are present in a typical diet and you'll find many referred to on the labels of processed foods.



Fatty acids have some important features which you should note:

- They have straight chains with no branching.
- They have even numbers of carbon atoms.
- They may be saturated with no double bonds in the chain or they may have one or more C=C double bonds in the chain, in which case they are usually *cis* (Z) alkenes. If there is more than one C=C double bond, they are not conjugated (either with the CO₂H group or with each other)—there is normally one saturated carbon atom between them.

Palmitic acid (C₁₆ saturated) is the most common fatty acid in living things. Oleic acid (C₁₈ mono-unsaturated) is the major fatty acid in olive oil. Arachidonic acid (C₂₀ tetra-unsaturated)

is a rare fatty acid, which is the precursor of the very important biological messengers the prostaglandins, thromboxanes, and leukotrienes.

The prevalence of fatty acids with even numbers of carbon atoms suggests a two-carbon building block, the most obvious being acetate. If labelled acetate is fed to plants, the fatty acids emerge with labels on alternate carbons like this.



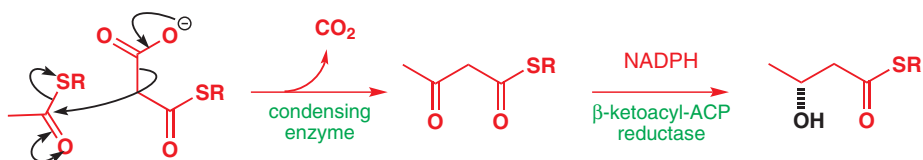
The green blob might represent deuterium (as a CD_3 group) and the black blob ^{13}C . In fact, the reactions are more complex than this suggests as CO_2 is also needed as well as CoA and it turns out that only the first two-carbon unit is put in as acetyl CoA. The remainder are added as malonyl CoA. If labelled malonyl CoA is fed, the starter unit, as it is called, is not labelled.

Malonyl CoA

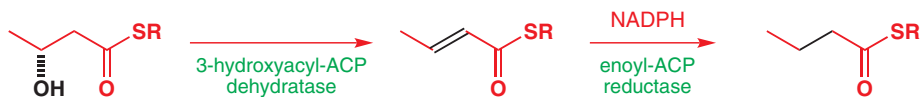
Malonyl CoA is the thiol ester of CoASH and malonic acid. It is biosynthesized by acylation of acetyl CoA with carbon dioxide.



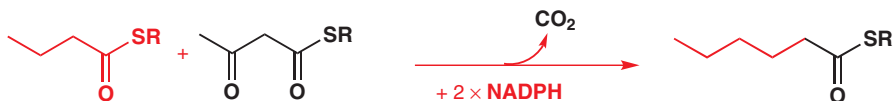
The first stage in fatty acid biosynthesis is a condensation between acetyl CoA (the starter unit) and malonyl CoA with the loss of CO_2 . This reaction could be drawn like this, with CO_2 being lost as the new C–C bond is formed. When chemists use malonates, we like to make the stable enol using both carbonyl groups, condense, and only afterwards release CO_2 (Chapter 25). As you saw on p. 1158, nature does this in making acetoacetyl CoA during alkaloid biosynthesis, but here things work differently.



The next step is reduction of the ketone group. This NADPH reaction is typically stereo- and chemoselective, although the stereochemistry is rather wasted here as the next step is a dehydration, typical of what is now an aldol product, and occurring by an enzyme-catalysed E1cB mechanism. The elimination is known to be a *cis* removal of H and OH, and the double bond is exclusively *trans* (*E*). Finally in this cycle, the double bond is reduced using another molecule of NADPH to give the saturated side chain.



Now the whole cycle can start again using this newly made C_4 fatty acid as the starter unit and building a C_6 fatty acid and so on. Each time the cycle turns, two carbon atoms are added to the acyl end of the growing chain.

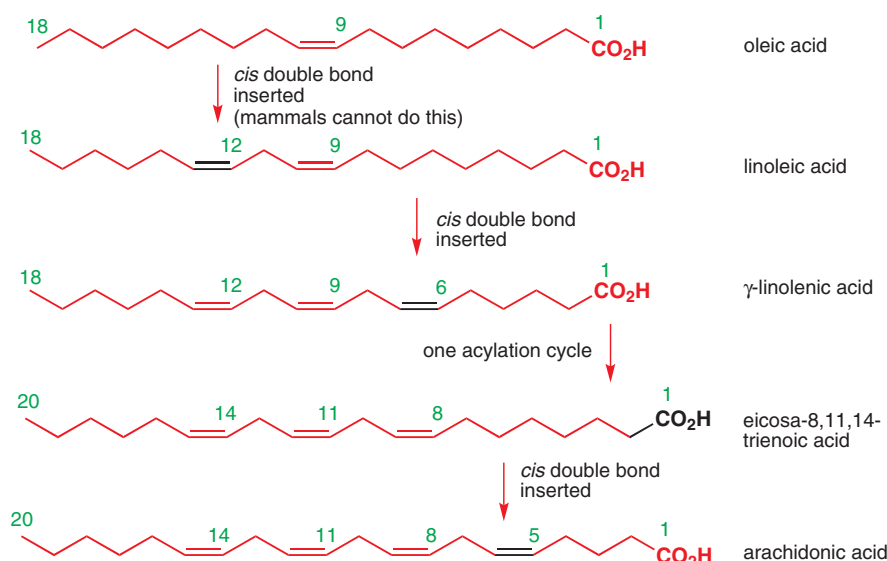


What is so important about unsaturated fatty acids?

Mammals can insert a *cis* alkene into the chain, providing that it is no further away from the carbonyl group than C9. We cannot synthesize linoleic or linolenic acids (see chart on p. 1161) directly as they have alkenes at C12 and C15, so these acids must be present in our diet.

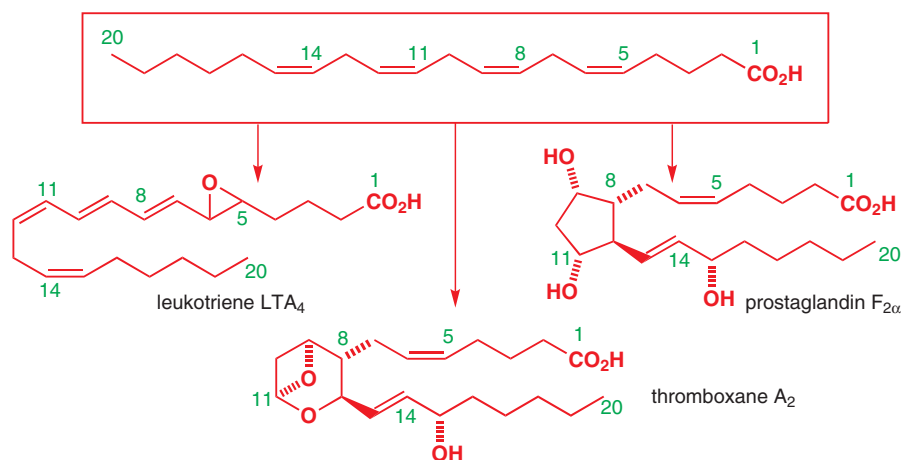
But why are we so keen to have them? They are needed for the synthesis of arachidonic acid, a C_{20} tetraenoic acid that is the precursor for some very interesting and important compounds. This is the biosynthesis of arachidonic acid.

biosynthesis of unsaturated fatty acids



The final product of this chain of events—arachidonic acid—is one of the eicosanoids, so-called because *eicosa* is Greek for twenty. The leukotrienes resemble arachidonic acid most closely, the prostaglandins have a closed chain forming a five-membered ring, and the thromboxanes resemble the prostaglandins but have a broken chain. All are C_{20} compounds with the sites of the alkenes (C5, C8, C11, and C14) marked by functionality or some other structural feature.

compounds synthesized from arachidonic acid



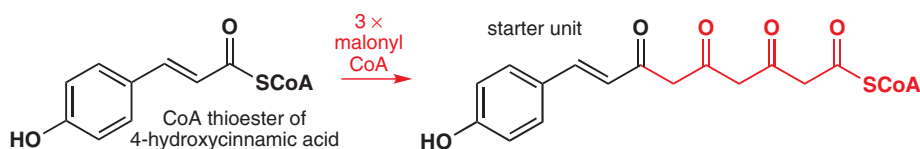
These compounds, made by oxidation of arachidonic acid, are all unstable and all are involved in transient events such as inflammation, blood clotting, fertilization, and immune responses. They are produced locally and decay quickly, and are implicated in autoimmune diseases like asthma and arthritis.

Aromatic polyketides

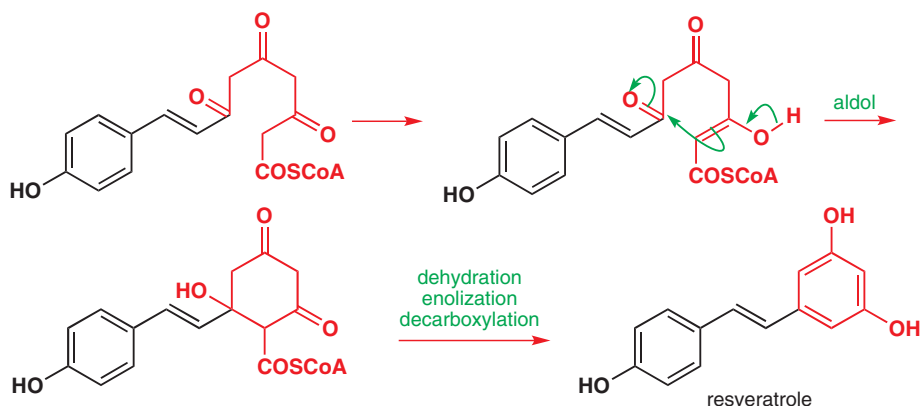
Other starter units such as 4-hydroxycinnamic acid, made from shikimic acid, can be used to build up aromatic compounds. The addition of three malonyl CoA units gives a linear

Full details of the biosynthesis of the fatty acids and their metabolites are in the online chapter 'Natural products'.

tetraketone (hence the name of this class of natural product) that can cyclize to resveratrole, a compound in red grape skins that has been suggested as one of the compounds in red wine that protects against heart disease.

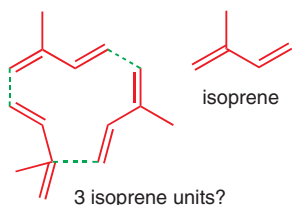
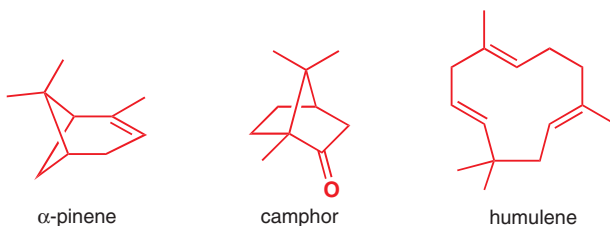


Redrawing this intermediate shows how easily it can cyclize to a six-membered ring. Enol formation allows a very favourable aldol cyclization to give a six-membered ring then dehydration and enolization to make the aromatic ring with hydrolysis of the CoA ester and decarboxylation gives resveratrole.



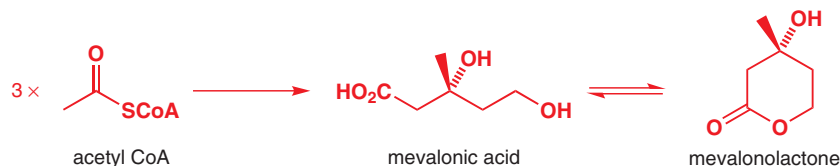
Terpenes are volatile constituents of plants

Terpenes were originally named after turpentine, the volatile oil from pine trees used in oil painting, whose major constituent is α -pinene. The term was rather vaguely used for all the volatile oily compounds, insoluble in water and usually with resinous smells from plants. Oils distilled from plants, which often contain perfumery or flavouring materials, are called essential oils and these too contain terpenes. Examples include camphor from the camphor tree, which is used to preserve clothes from moths, and humulene from hops, which helps to give beer its flavour.

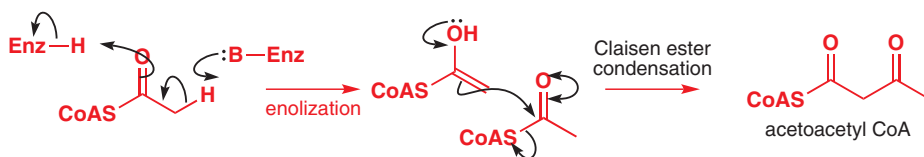


You will notice that they are all aliphatic compounds with a scattering of double bonds and rings, few functional groups, and an abundance of methyl groups. A better definition (that is, a biosynthetically based definition) arose when it was noticed that all these compounds have $5n$ carbon atoms. Pinene and camphor are C_{10} compounds while humulene is C_{15} . It seemed obvious that terpenes were made from a C_5 precursor and the favourite candidate was isoprene (2-methylbuta-1,3-diene) as all these structures can be drawn by joining together 2-, 3-, or 4-isoprene skeletons end to end.

In fact, this is not correct. Isoprene is not an intermediate, and the discovery of the true pathway started when acetate was, rather surprisingly, found to be the original precursor for all terpenes. The key intermediate is mevalonic acid, formed from three acetate units and usually isolated as its lactone.



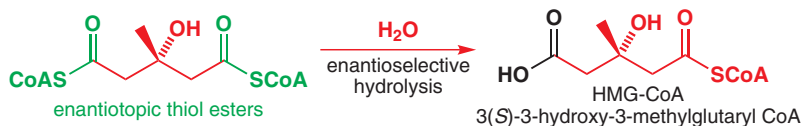
The first step is the Claisen ester condensation of two molecules of acetyl CoA, one acting as an enol and the other as an electrophilic acylating agent to give acetoacetyl CoA. We saw the same reaction in the biosynthesis of the pyrrolidine alkaloids earlier in this chapter.



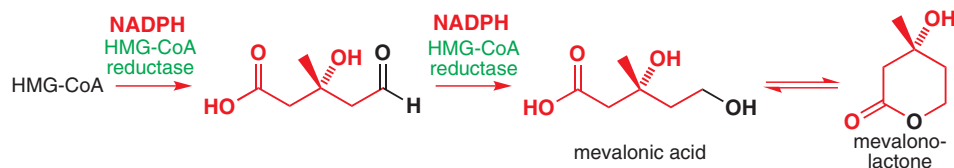
The third molecule of acetyl CoA also functions as a nucleophilic enol and attacks the keto group of acetoacetyl CoA. This is not a Claisen ester condensation—it is an aldol reaction between the enol of a thiol ester and an electrophilic ketone.



We have drawn the product with stereochemistry even though it is not chiral. This is because one of the two enantiotopic thiol esters is hydrolysed while this intermediate is still bound to the enzyme, so a single enantiomer of the half-acid/half-thiol ester results.

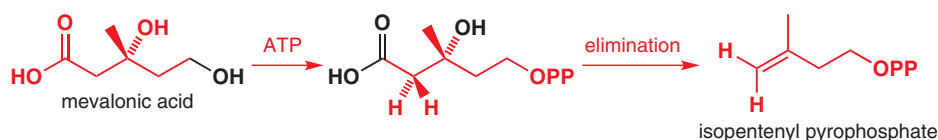
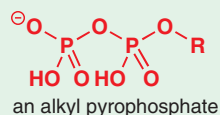


The remaining thiol ester is more electrophilic than the acid and can be reduced by the nucleophilic hydride from NADPH. Just as in LiBH_4 reductions of esters (Chapter 23), the reaction does not stop at the aldehyde level, and two molecules of NADPH are used to make the alcohol. This is mevalonic acid.

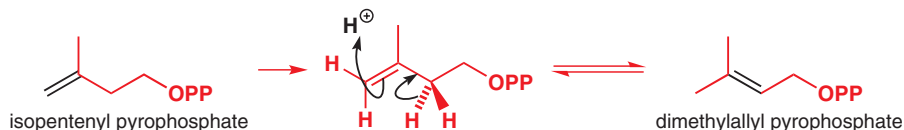


Mevalonic acid is indeed the true precursor of the terpenes but it is a C_6 compound and so it must lose a carbon atom to give the C_5 precursor. The spare carbon atom becomes CO_2 by an elimination reaction. First, the primary alcohol is pyrophosphorylated with ATP; then the CO_2H group and the tertiary alcohol are lost in a concerted elimination.

■ 'PP' indicates the pyrophosphate group transferred from ATP.



So is isopentenyl pyrophosphate the C_5 intermediate at last? Well, yes and no. There are actually two closely related C_5 intermediates, each of which has a specific and appropriate role in terpene biosynthesis. Isopentenyl pyrophosphate is in equilibrium with dimethylallyl pyrophosphate by a simple allylic proton transfer.



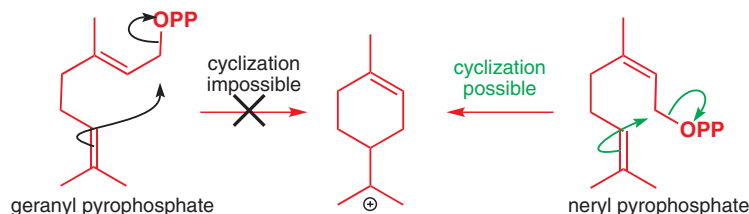
The two C_5 intermediates now react with each other. The dimethylallyl pyrophosphate is the better electrophile because it is allylic, and allylic compounds are good at both S_N1 and S_N2 reactions (Chapter 15). Isopentenyl pyrophosphate is the better nucleophile because it can react through an unhindered primary carbon atom to produce a tertiary cation—we can draw the reaction like this:



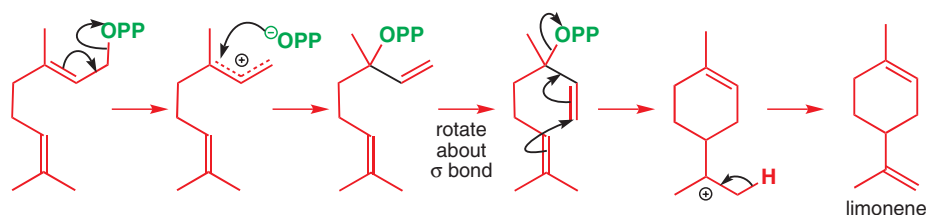
Although this idea reveals the thinking behind the reaction, in fact it does not go quite like this. The product is one particular positional and geometrical isomer of an alkene and the cation is not an intermediate. Indeed, the reaction is also stereospecific (discovered again by proton labelling, but we will not give the rather complex details) and this too suggests a concerted process.



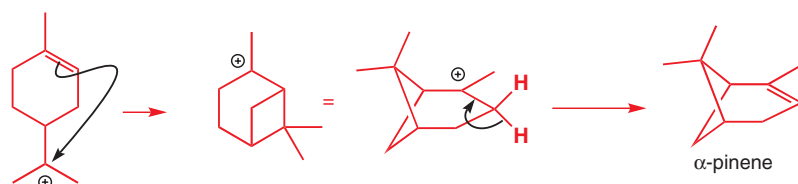
As soon as we start to make typical cyclic monoterpenes from geranyl pyrophosphate we run into a snag. We cannot cyclize geranyl pyrophosphate because it has a *trans* double bond! We *could* cyclize the *cis* compound (neryl pyrophosphate), and it used to be thought that this was formed from the *trans* compound as an intermediate.



It is now known that nature gets round this problem without making neryl pyrophosphate. An allylic rearrangement occurs to move the pyrophosphate group to the tertiary centre. This is an unfavourable rearrangement thermodynamically and probably occurs via the allyl cation and is catalysed by $Mg(II)$. There is no longer any geometry about the alkene. The molecule can now rotate freely about a single bond and cyclization can occur. Even if only a small amount of the rearranged allylic pyrophosphate is present, that can rearrange and more can isomerize.

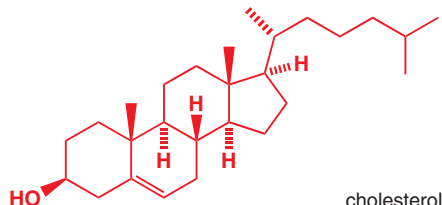
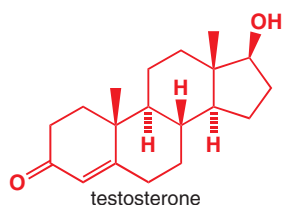


More interesting compounds come from the cyclization of the first formed cation. The remaining alkene can attack the cation to form what looks at first to be a very unstable compound but which is actually a tertiary carbocation with the pinene skeleton. There are many thousands of terpenes with multiple C_5 units all made from mevalonic acid.



The steroids are another group of compounds derived from mevalonic acid. They include sex hormones such as testosterone and progesterone, and the cholesterol needed to build cell membranes but also implicated in the damage to arteries caused by atherosclerosis.

The synthesis of the steroids is discussed in the online chapter 'Natural products'.



The elucidation of the ways in which organic chemistry underpins life, along with the use of organic chemistry to construct in the laboratory the molecules used by nature, has been one of the greatest scientific success stories of recent decades. In this chapter we have revealed but a glimpse of the immense complexity of the world of biological organic chemistry; you will find an extended version of this discussion in the three chapters on the web, and a book on biochemistry will fill in more detail. The beautiful molecular structures of nature and the reactions used to make them have provided an example for organic chemists to follow—sometimes at a distance, but always in hot pursuit. The next and final chapter of this book tells a few stories of how such scientific inspiration is the key to the future of chemistry, not only for its own sake, but also for the sake of the millions of people whose lives have been improved or even saved by the ingenuity of chemists.

Further reading

P. A. Frey and A. D. Hegeman, *Enzymatic Reaction Mechanisms*, Oxford University Press, Oxford, 2007. A more basic treatment is in two Oxford Primers by J. Mann, *Chemical Aspects of Biosynthesis*, OUP, 1994 and by T. Bugg, *Introduction to Enzyme and Coenzyme Chemistry*, OUP, Oxford, 2004. A more comprehensive treatment is in J. E. McMurry and T. P. Begley, *The Organic Chemistry of Biological Pathways*, Roberts, 2005. For an introduction to biosynthesis, see F. J. Leeper and J. C. Vederas, *Biosynthesis: Polyketides and Vitamins*, Springer, 2000.

Three full chapters from the first edition expand this chapter and are available for download from the website: The chemistry of life, Mechanisms in biological chemistry, and Natural products.

Check your understanding



To check that you have mastered the concepts presented in this chapter, attempt the problems that are available in the book's Online Resource Centre at <http://www.oxfordtextbooks.co.uk/orc/clayden2e/>